



US012411569B2

(12) **United States Patent**
Bok et al.

(10) **Patent No.:** **US 12,411,569 B2**
(45) **Date of Patent:** ***Sep. 9, 2025**

(54) **DISPLAY DEVICE**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **18/530,598**

(22) Filed: **Dec. 6, 2023**

(65) **Prior Publication Data**

US 2024/0103653 A1 Mar. 28, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/986,718, filed on
Nov. 14, 2022, now Pat. No. 11,861,092, which is a
(Continued)

(30) **Foreign Application Priority Data**

Mar. 21, 2016 (KR) 10-2016-0033302

(51) **Int. Cl.**
G06F 3/041 (2006.01)
G02F 1/1333 (2006.01)

(Continued)

(52) **U.S. Cl.**
CPC **G06F 3/0412** (2013.01); **G02F 1/13338**
(2013.01); **G06F 3/0443** (2019.05);
(Continued)

(58) **Field of Classification Search**

CPC G06F 3/0412; G06F 3/0443; G06F 3/0445;
G06F 3/0446; G06F 2203/04105;
(Continued)

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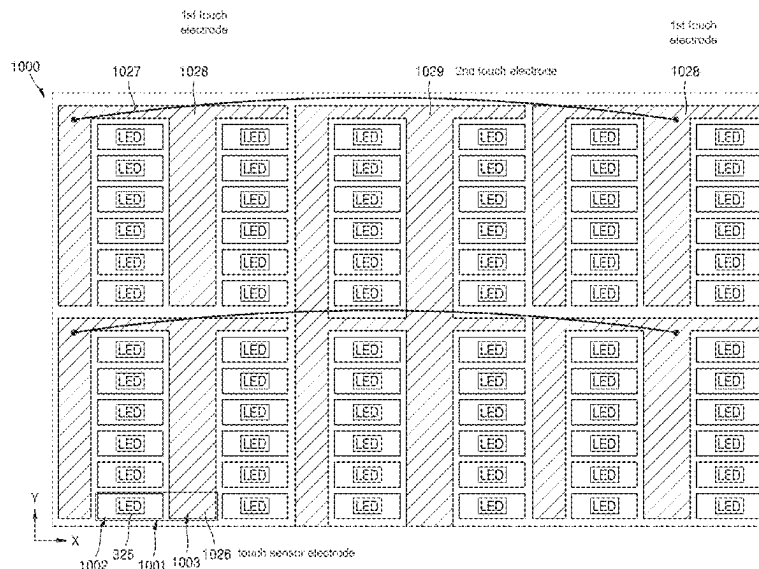
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(57) **ABSTRACT**

A display device includes a display substrate in which a
plurality of sub-pixel areas is defined, a light-emitting diode
("LED") on the display substrate, a touch sensor electrode
on the display substrate and including at least one touch
electrode, and a bank separating the plurality of the sub-
pixel areas, where each of the plurality of the sub-pixel areas
may include a first area in which the LED is disposed and a
second area in which the touch sensor electrode is disposed.

20 Claims, 17 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/420,660, filed on May 23, 2019, now Pat. No. 11,500,484, which is a continuation of application No. 15/373,877, filed on Dec. 9, 2016, now Pat. No. 10,338,712.

(51) Int. Cl.

G06F 3/044 (2006.01)
H10D 86/40 (2025.01)
H10D 86/60 (2025.01)
H10H 29/14 (2025.01)
H10K 59/121 (2023.01)
H10K 59/122 (2023.01)
H10K 59/123 (2023.01)
H10K 59/124 (2023.01)
H10K 59/40 (2023.01)

(52) U.S. Cl.

CPC **G06F 3/0445** (2019.05); **G06F 3/0446** (2019.05); **H10D 86/451** (2025.01); **H10D 86/60** (2025.01); **H10H 29/142** (2025.01); **H10K 59/1213** (2023.02); **H10K 59/122** (2023.02); **H10K 59/123** (2023.02); **H10K 59/124** (2023.02); **H10K 59/40** (2023.02); **G06F 2203/04105** (2013.01); **G06F 2203/04111** (2013.01)

(58) Field of Classification Search

CPC **G06F 2203/04111**; **G06F 3/044**; **G02F 1/13338**; **H01L 27/1248**; **H01L 27/156**; **H10K 59/1213**; **H10K 59/122**; **H10K 59/123**; **H10K 59/124**; **H10K 59/40**; **G09F 9/33**; **G09G 3/32**

See application file for complete search history.

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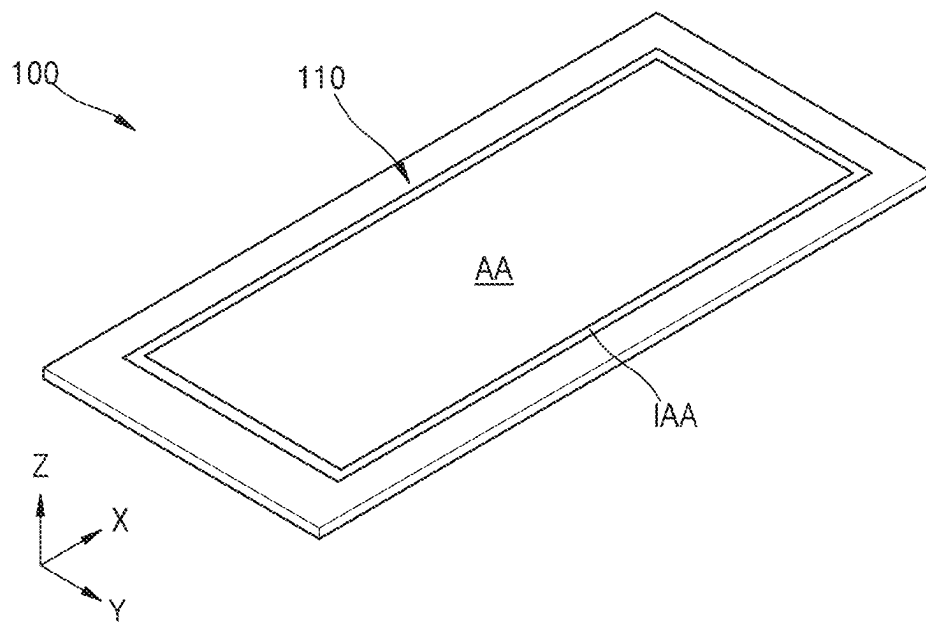
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FIG. 1



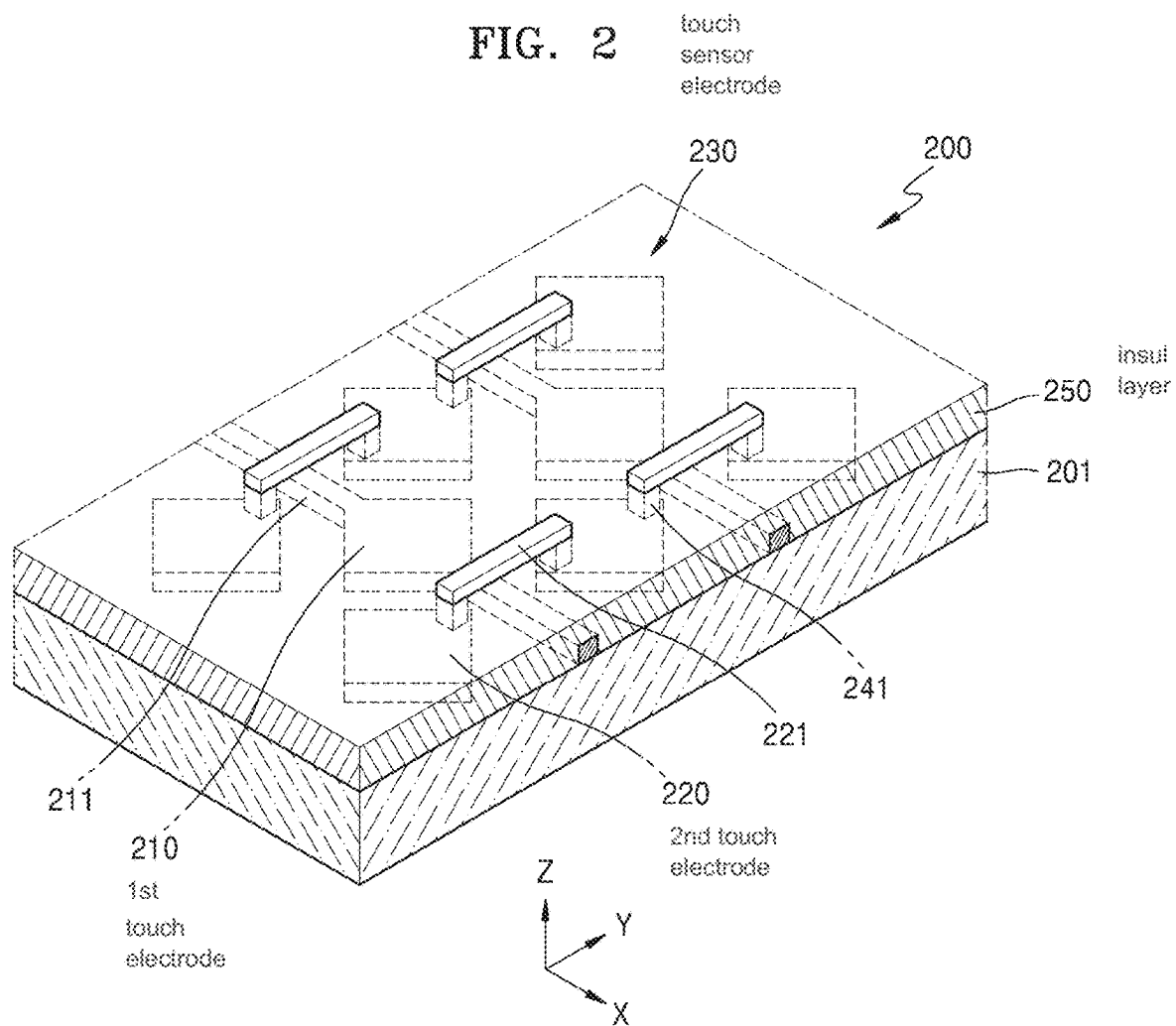


FIG. 3

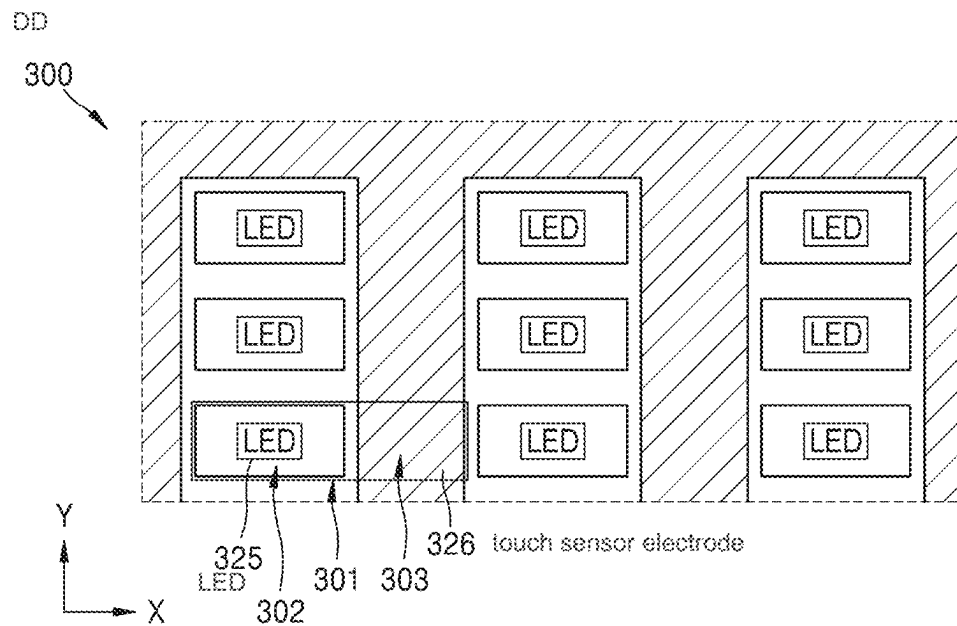


FIG. 4

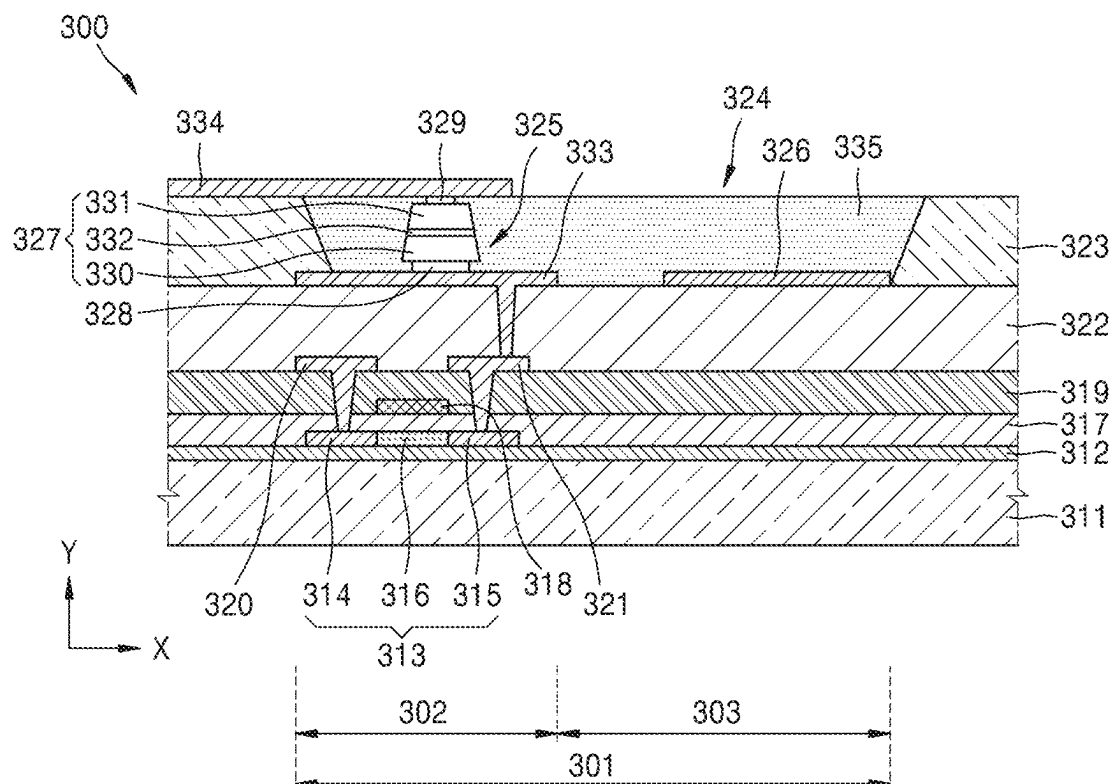


FIG. 5

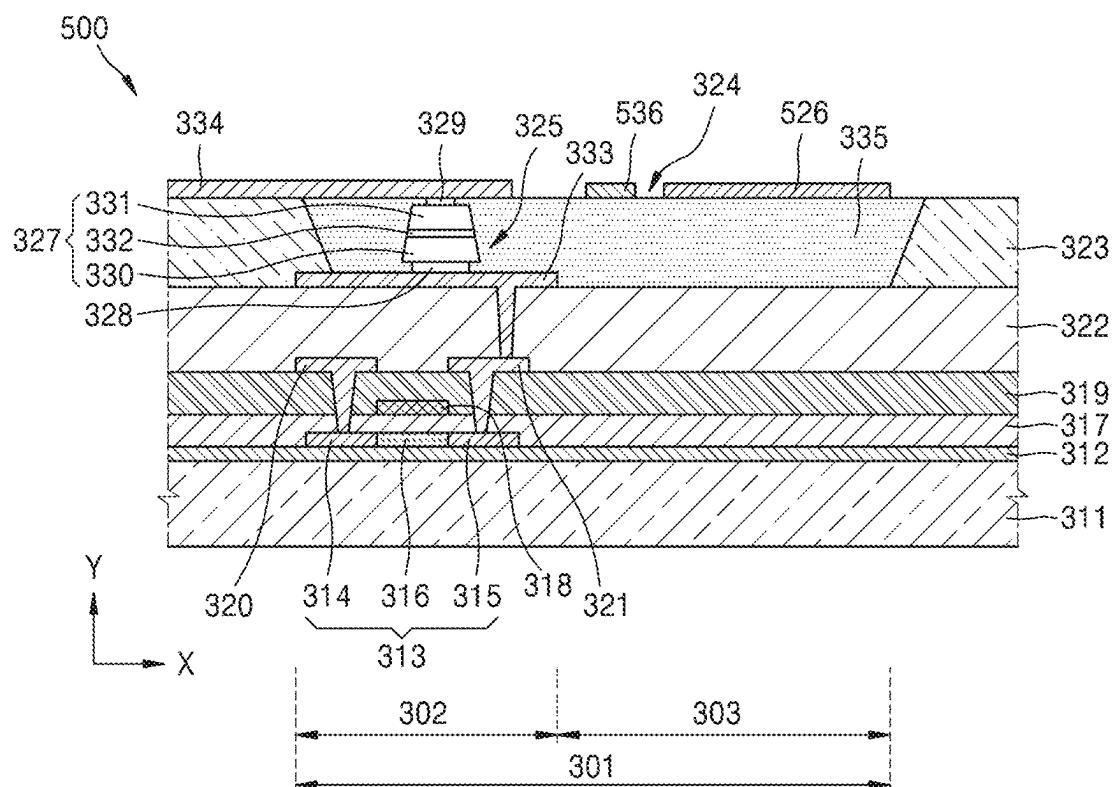


FIG. 6

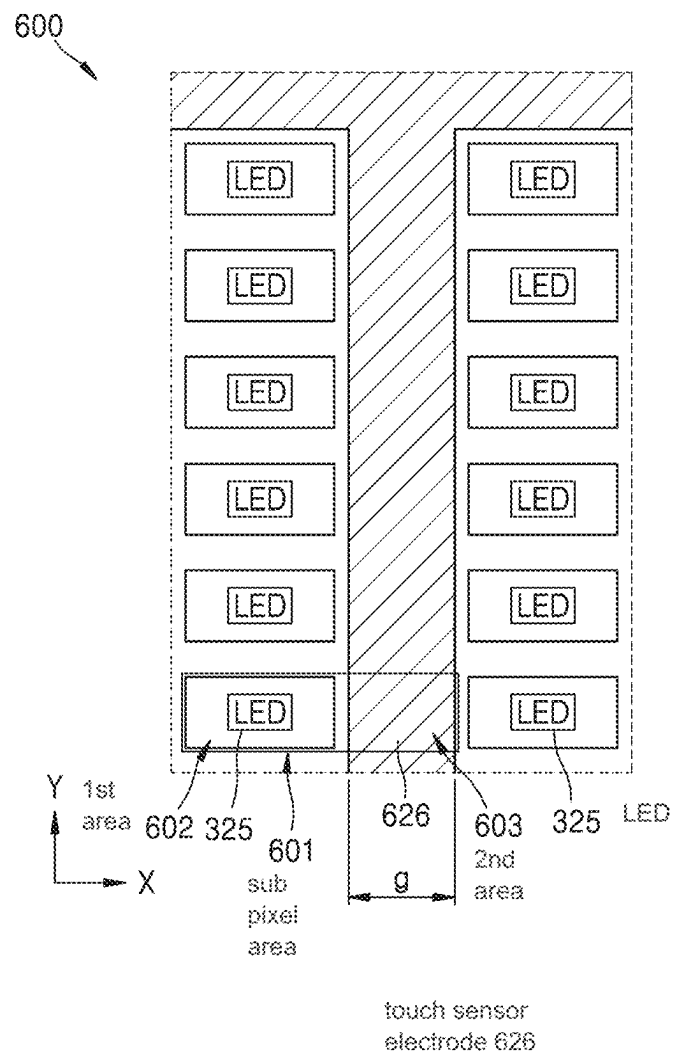


FIG. 7

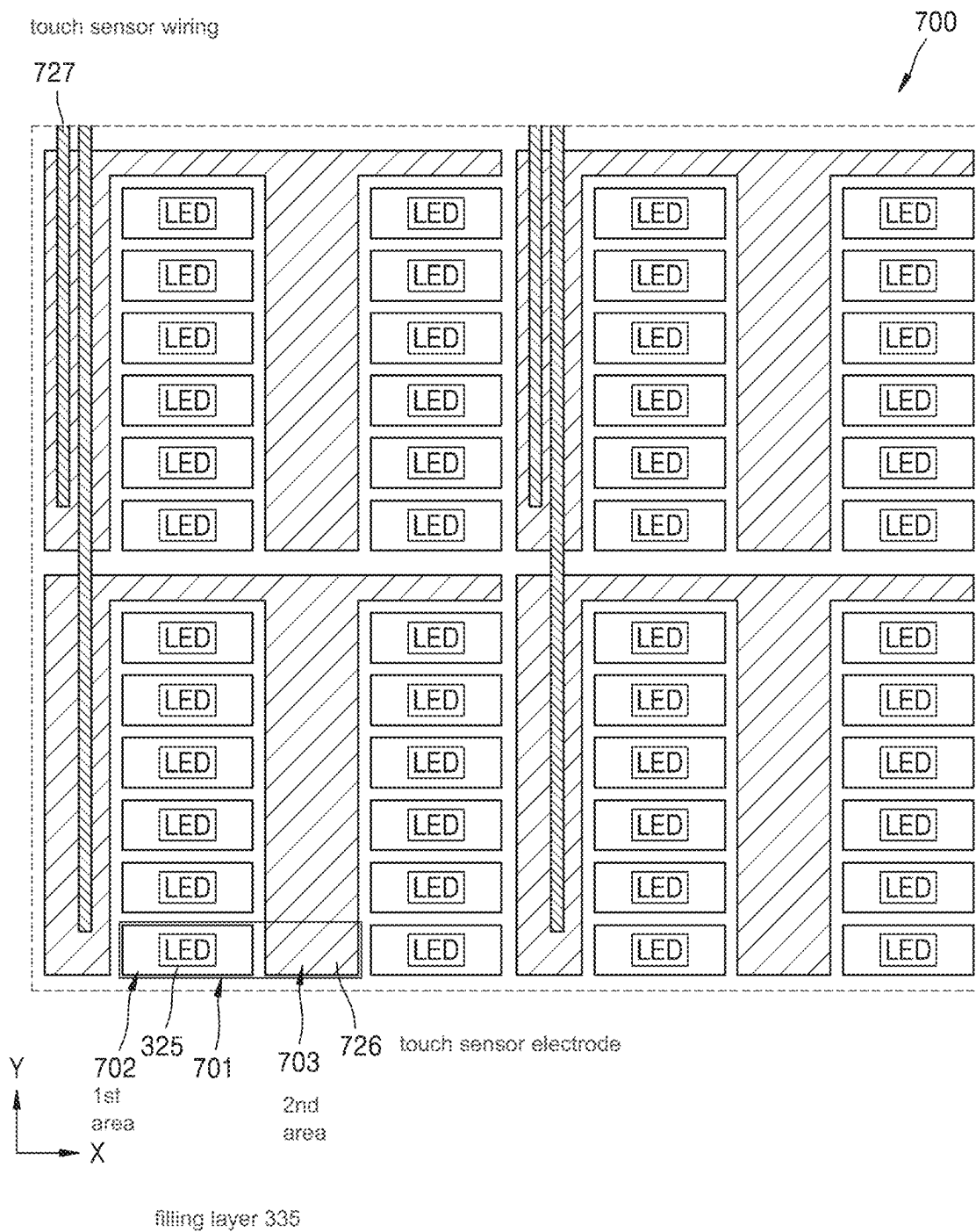


FIG. 8

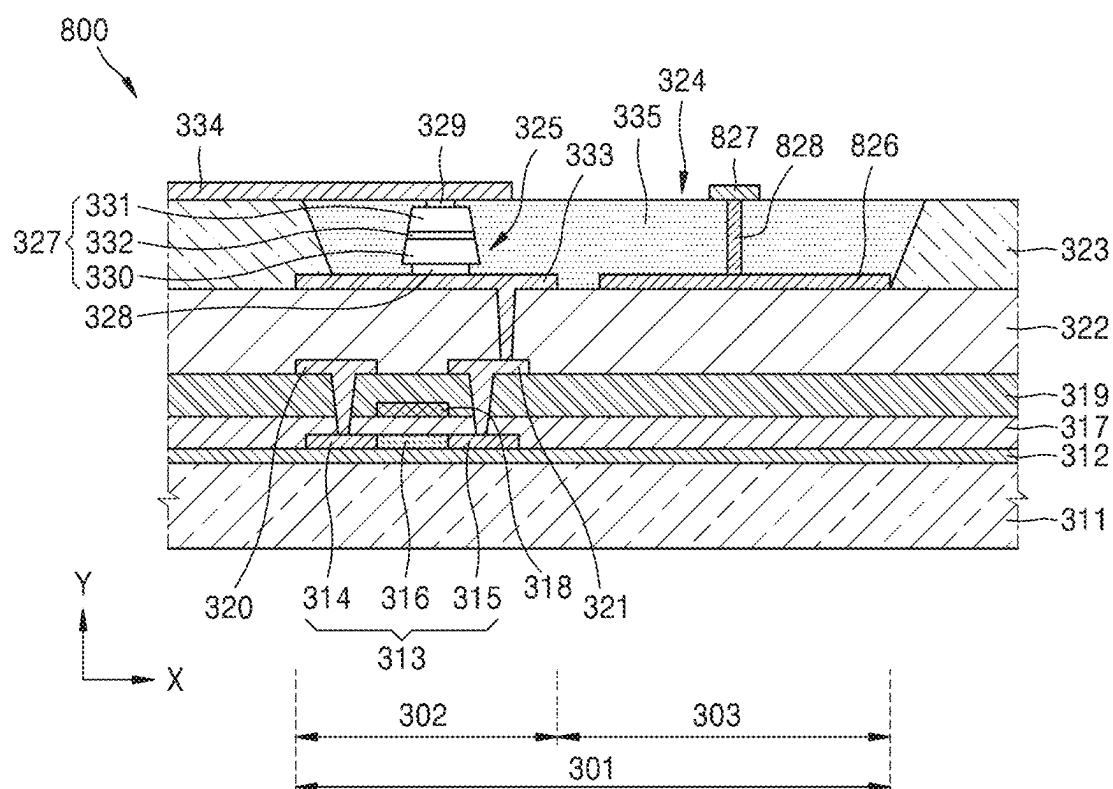
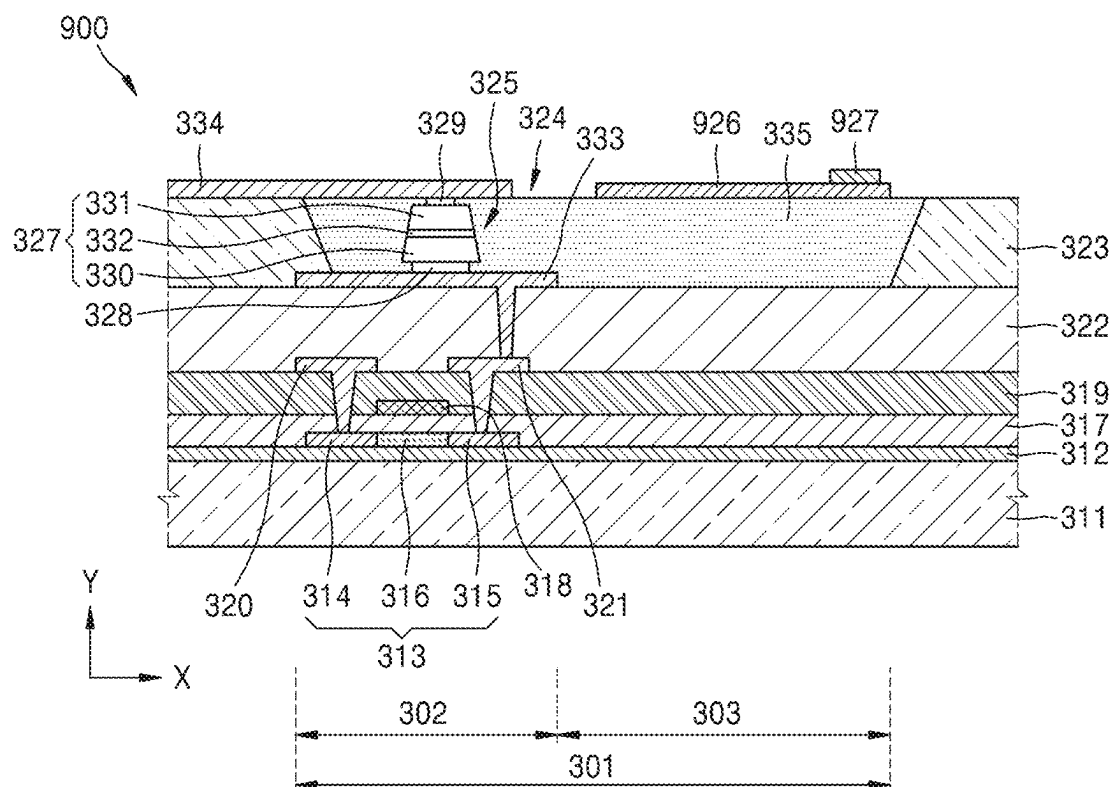
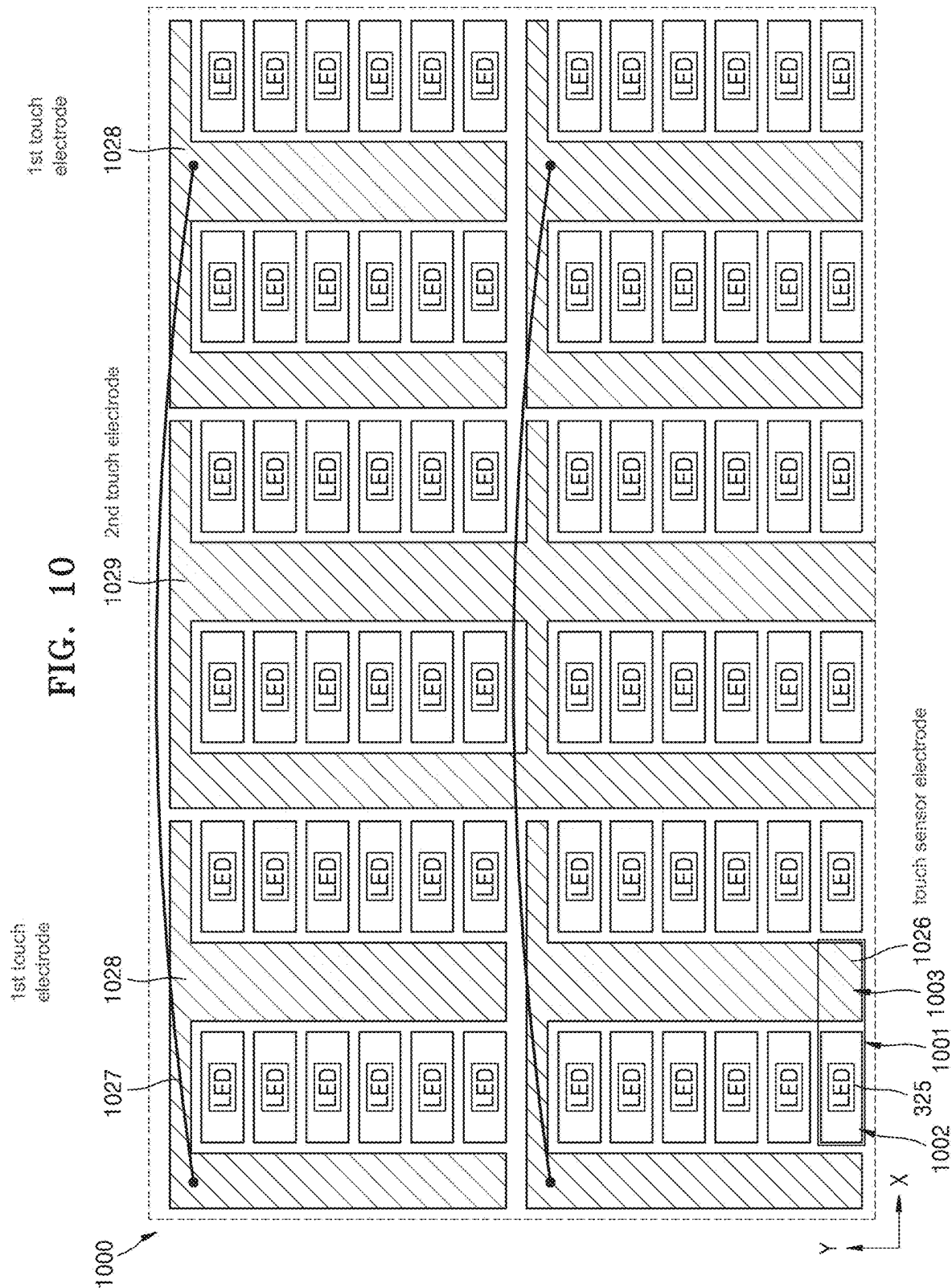


FIG. 9



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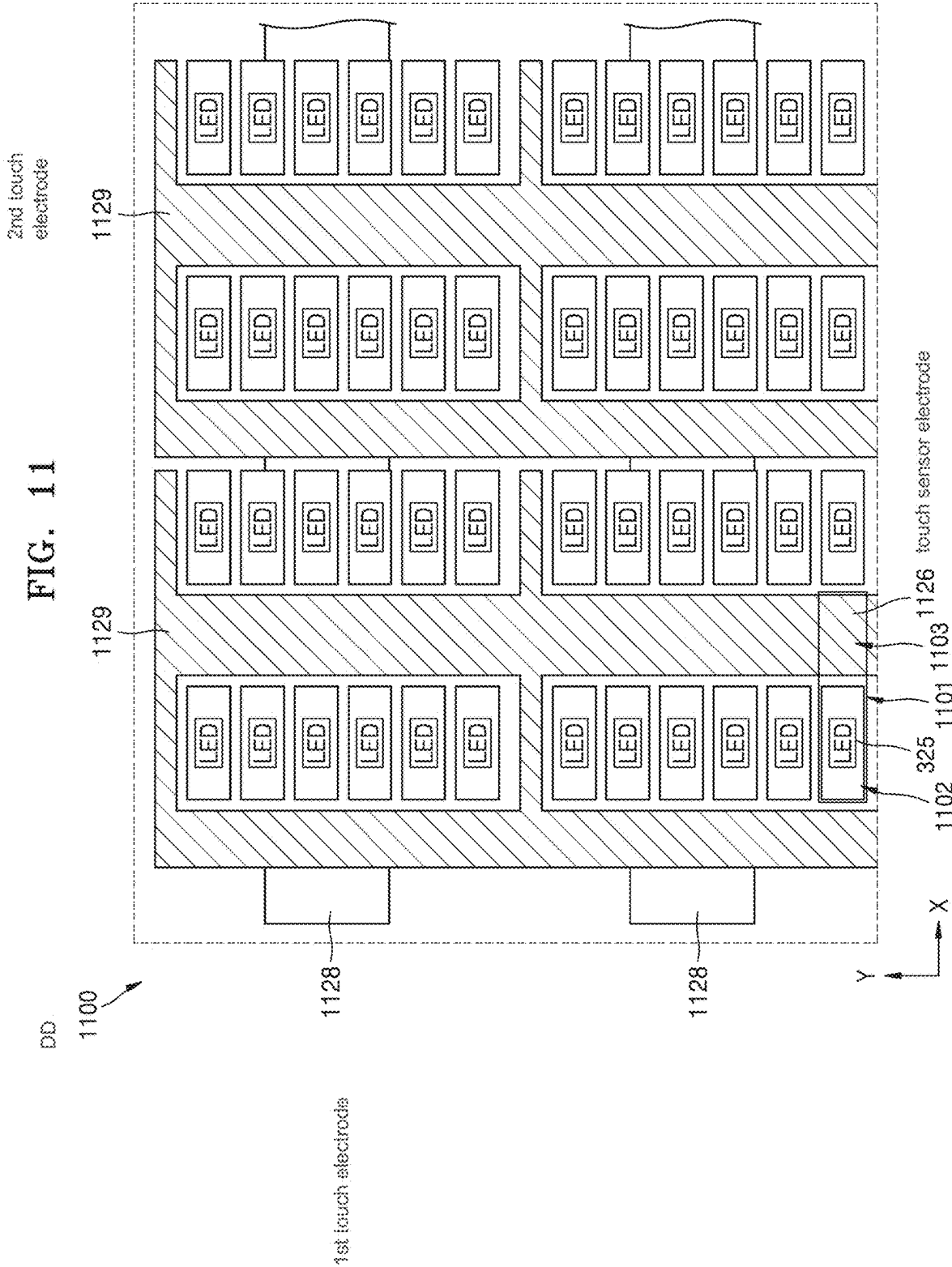


FIG. 12

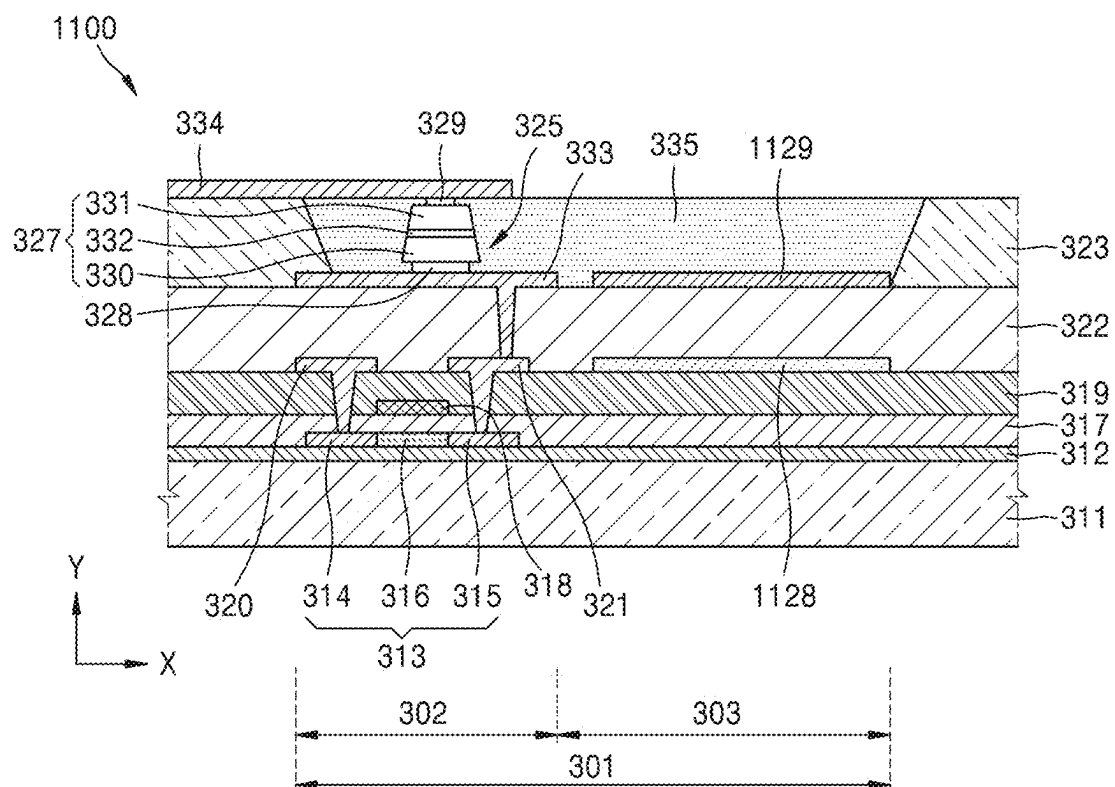


FIG. 13

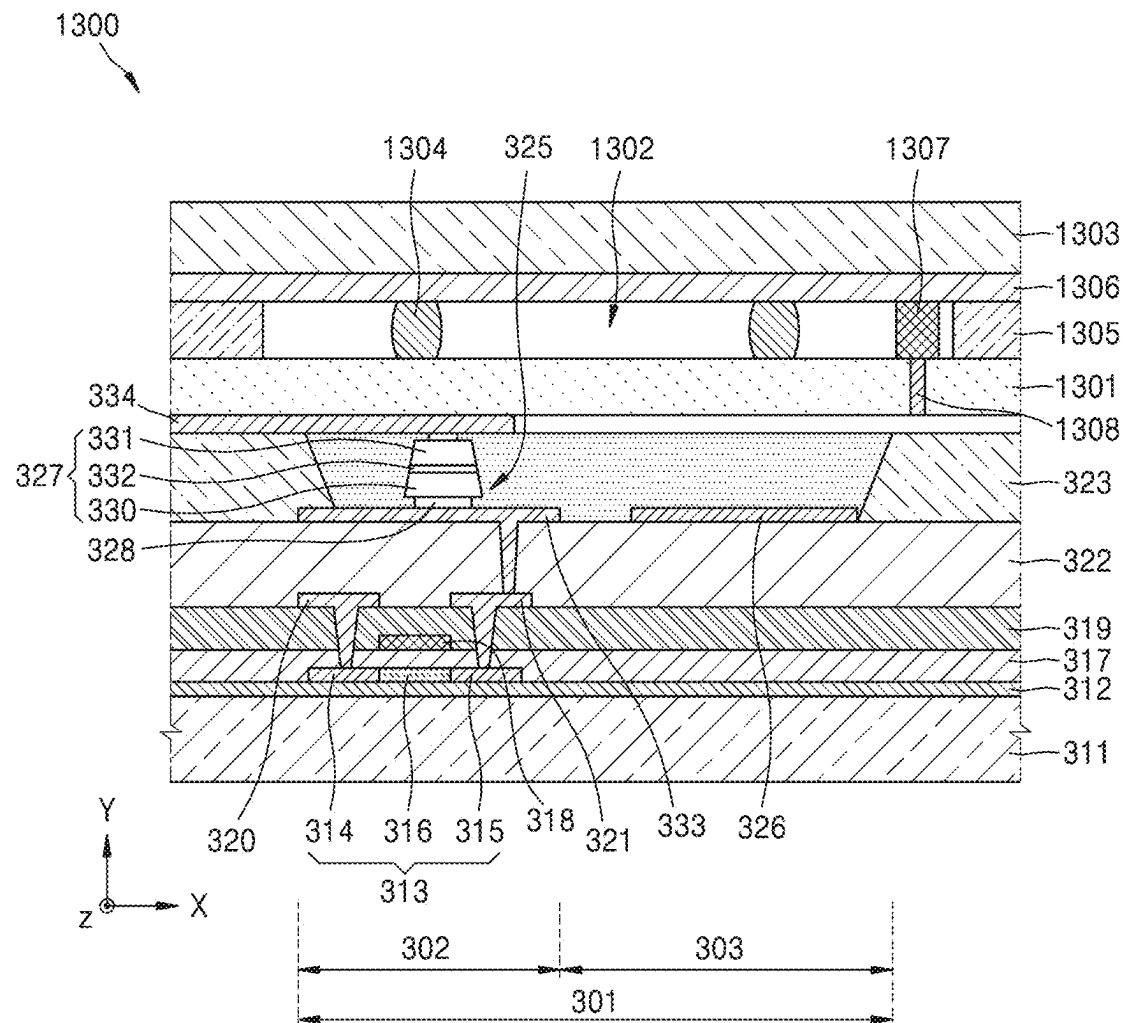


FIG. 14

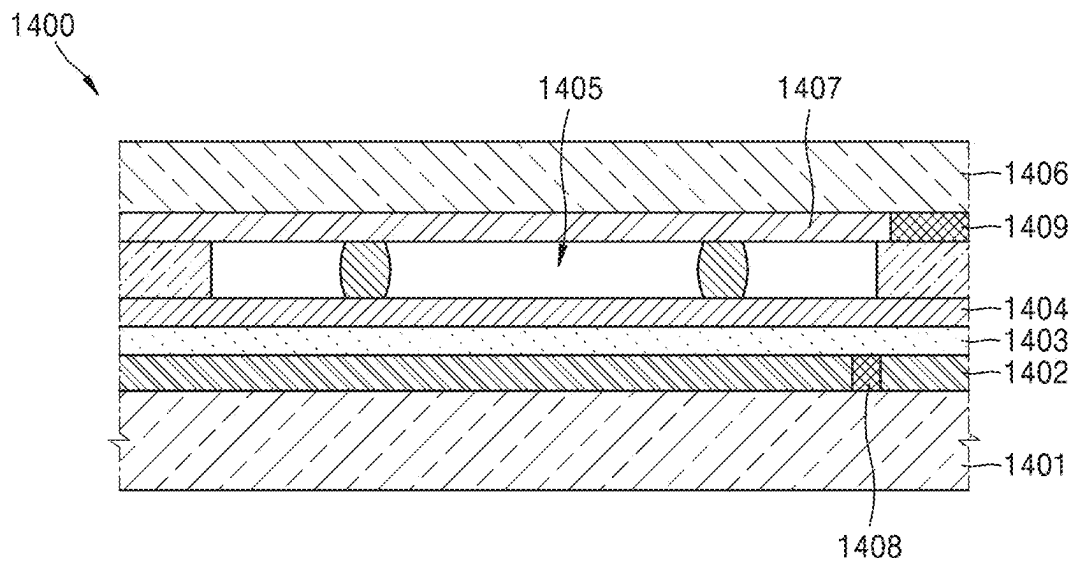


FIG. 15

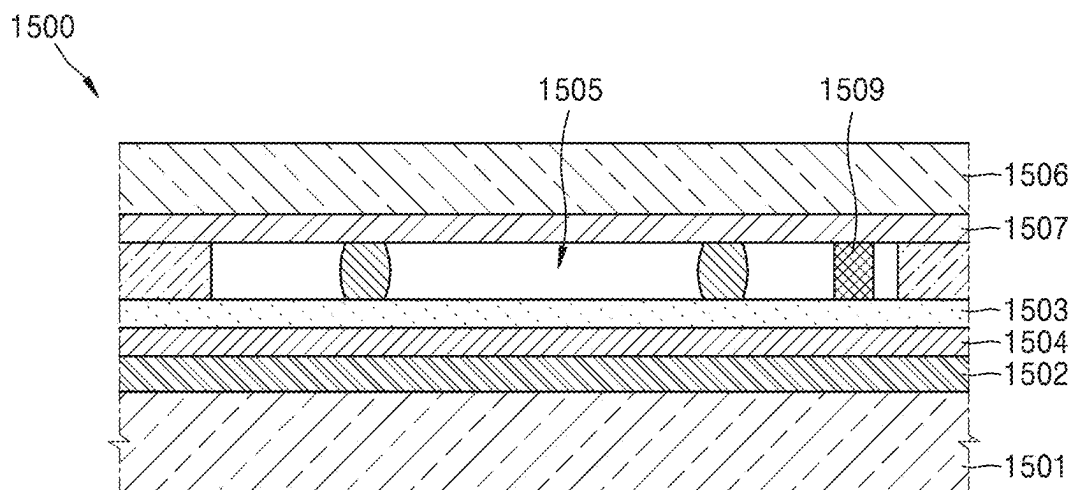


FIG. 16

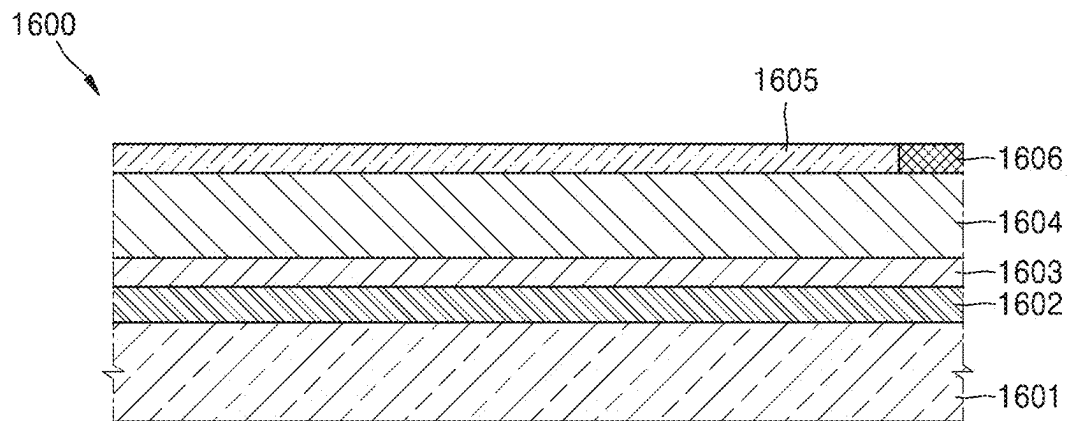


FIG. 17

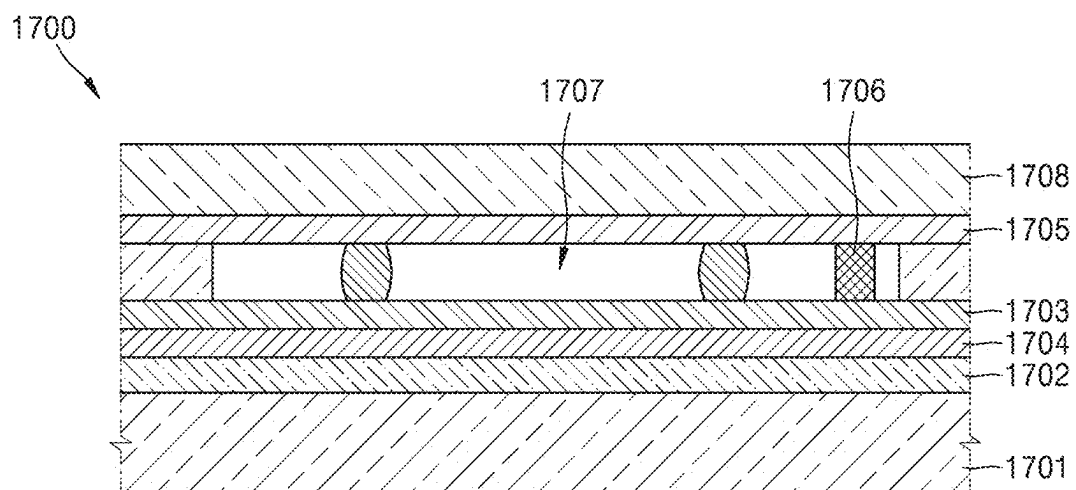


FIG. 18

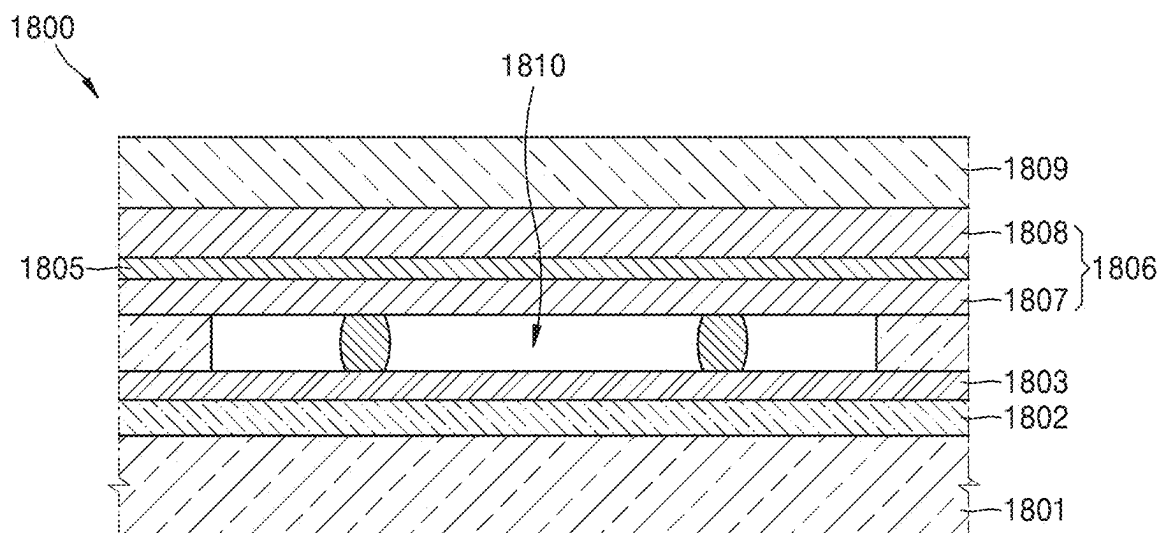
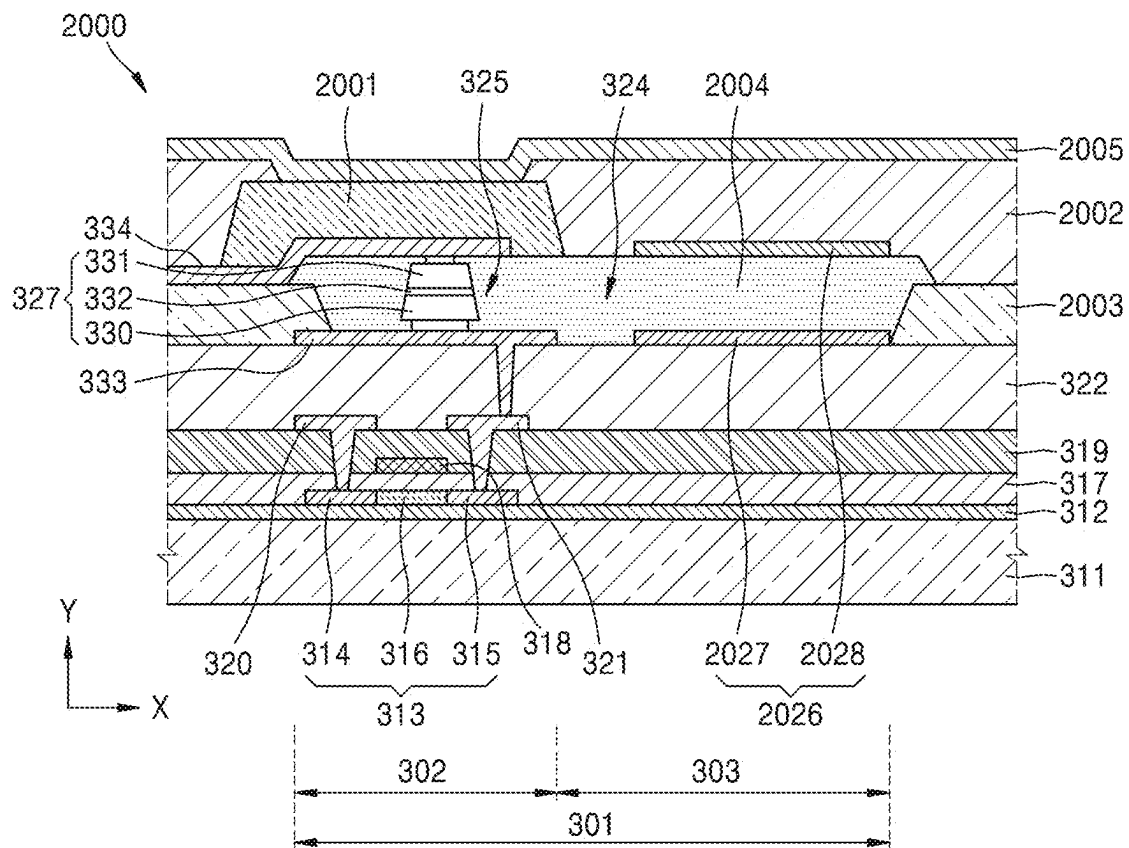


FIG. 20



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DISPLAY DEVICE

This application is a continuation of U.S. patent application Ser. No. 17/986,718, filed on Nov. 14, 2022, which is a continuation of U.S. patent application Ser. No. 16/420,660, filed on May 23, 2019, which is a continuation of U.S. patent application Ser. No. 15/373,877, filed on Dec. 9, 2016, which claims priority to Korean Patent Application No. 10-2016-0033302, filed on Mar. 21, 2016, and all the benefits accruing therefrom under 35 U.S.C. § 119, the content of which in its entirety is herein incorporated by reference.

BACKGROUND

1. Field

One or more exemplary embodiments relate to a display device.

2. Description of the Related Art

A light-emitting diode (“LED”) is one type of a semiconductor element, that is, a p-n junction diode, in which holes and electrons are injected when a forward bias is applied thereto and energy generated by recombination of the holes and electrons is transformed into light.

The LED may be used in various types of mobile devices such as a smart phone, a lap top computer, a digital camera, a camcorder, a mobile information terminal, a tablet personal computer and a watch, and in various types of electronic devices such as a desk top computer, a television (“TV”) set, an outdoor advertisement board, an exhibition display device, an automobile instrument panel and a head up display (“HUD”).

SUMMARY

One or more exemplary embodiments include a display device which displays images via light-emitting diodes (“LEDs”) and has touch functions that can be performed via a touch screen.

Additional exemplary embodiments will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented exemplary embodiments.

According to an exemplary embodiment of the invention, a display device may include a display substrate having a plurality of sub-pixel areas, an LED above the display substrate, a touch sensor electrode above the display substrate and including at least one touch electrode, and a bank separating the sub-pixel areas, where each of the plurality of the sub-pixel areas may include a first area in which the LED is disposed and a second area in which the touch sensor electrode is disposed.

According to one or more exemplary embodiments, each of the plurality of the sub-pixel areas may be positioned in an opening surrounded by the bank, and the first area in which the LED arranged is disposed and the second area in which the touch sensor electrode is disposed may be adjacent to each other in each sub-pixel area.

According to one or more exemplary embodiments, the LED may include a first contact electrode, a second contact electrode, and a p-n diode arranged between the first and second contact electrodes, where the first contact electrode may be arranged on a bottom portion of the LED and electrically connected to a first electrode connected to a thin

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film transistor, and the second contact electrode may be electrically connected to a second electrode arranged on a top portion of the LED.

According to one or more exemplary embodiments, the touch sensor electrode and the first electrode may be arranged on the same layer.

According to one or more exemplary embodiments, the touch sensor electrode and the second electrode may be arranged on the same layer.

According to one or more exemplary embodiments, a filling layer embedding the LED may fill the opening and the touch sensor electrode may be arranged on the filling layer.

According to one or more exemplary embodiments, the touch sensor electrode may include an electrode operating in a self-capacitance method and sense a capacitance change of a single touch electrode, and touch sensor wirings may be respectively and electrically connected to the touch sensor electrode and reciprocally exchange electrical signals of the touch sensor electrodes with an external device.

According to one or more exemplary embodiments, the touch sensor electrode and the first electrode may be arranged on the same layer, a filling layer may be arranged in the opening, the LED and the touch sensor electrode being embedded in the filling layer, and the touch sensor wiring electrically connected to the touch sensor electrode via a contact hole may be arranged on the filling layer.

According to one or more exemplary embodiments, the filling layer may be arranged in the opening, the LED being embedded in the filling layer, the touch sensor electrode may be arranged on the filling layer, the touch sensor electrode and the second electrode may be on the same layer, and the touch sensor wiring may be arranged on the touch sensor electrode.

According to one or more exemplary embodiments, the touch sensor electrode may include an electrode operating in a mutual-capacitance method and sense a capacitance change occurring between a first touch electrode and a second touch electrode.

According to one or more exemplary embodiments, the first and second touch electrodes and the first or second electrodes may be arranged on the same layer.

According to one or more exemplary embodiments, a plurality of first touch electrodes may be electrically connected via connecting wirings arranged on a different layer with respect to the first touch electrode, and the second touch electrode may be electrically connected to respective touch sensor wirings.

According to one or more exemplary embodiments, the first and second touch electrodes may be separated on different layers.

According to one or more exemplary embodiments, the first touch electrode and at least one of a gate electrode, a source electrode, and a drain electrode, which are provided in the thin film transistor, may be arranged on the same layer, and the second touch electrode and the first or second electrodes may be arranged on the same layer.

According to one or more exemplary embodiments, the touch sensor electrode may include an electrode which senses capacitance changes along an X-axis and a Y-axis crossing the X-axis, and a force sensing electrode which senses, together with the touch sensor electrode, a capacitance change along a Z-axis perpendicular to the X-axis and the Y-axis may be arranged on the display substrate.

According to one or more exemplary embodiments, at least one of function layers covering the sub-pixel area may

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be arranged above the display substrate and the force sensing electrode may be arranged above the at least one of the function layer.

According to one or more exemplary embodiments, the force sensing electrode may include a conductive polymer material.

According to one or more exemplary embodiments, a refractive index matching layer having a refractive index less than that of the force sensing electrode may be arranged on at least one surface of the force sensing electrode.

According to one or more exemplary embodiments, the bank may include a first bank and a second bank arranged on the first bank, the LED may be arranged in an opening surrounded by the first bank, the second bank surrounding the LED may be arranged on the first bank, and the second electrode may be electrically connected to the LED on the second bank.

According to one or more exemplary embodiments, the thin film transistor arranged on a bottom portion of the sub-pixel area may include a gate electrode, a source electrode, and a drain electrode, the first touch electrode and at least one of the first electrode, the gate electrode, the source electrode, and the drain electrode may be arranged on the same layer, and the second touch electrode and the second electrode may be arranged on the same layer.

BRIEF DESCRIPTION OF THE DRAWINGS

These and/or other exemplary embodiments will become apparent and more readily appreciated from the following description of the exemplary embodiments, taken in conjunction with the accompanying drawings in which:

FIG. 1 is a perspective view of an exemplary embodiment of a display device;

FIG. 2 is a partially exploded perspective view of an exemplary embodiment of a touch screen panel;

FIG. 3 is a plan view of the display device, showing an exemplary embodiment of an arrangement of a light-emitting diode ("LED") and a touch sensor electrode;

FIG. 4 is a cross-sectional view of a single sub-pixel in FIG. 3;

FIG. 5 is a plan view of another exemplary embodiment of a display device;

FIG. 6 is a plan view of an exemplary embodiment of an arrangement of an LED and a touch sensor electrode;

FIG. 7 is a plan view of another exemplary embodiment of an arrangement of an LED and a touch sensor electrode.

FIG. 8 is a cross-sectional view of another exemplary embodiment of a single sub-pixel;

FIG. 9 is a cross-sectional view of another exemplary embodiment of a single sub-pixel;

FIG. 10 is a plan view of another exemplary embodiment of an arrangement of an LED and a touch sensor electrode;

FIG. 11 is a plan view of another exemplary embodiment of an arrangement of an LED and a touch sensor electrode.

FIG. 12 is a cross-sectional view of a single sub-pixel in FIG. 11;

FIG. 13 is a cross-sectional view of another exemplary embodiment of a display device;

FIG. 14 is a cross-sectional view of another exemplary embodiment of a display device;

FIG. 15 is a cross-sectional view of another exemplary embodiment of a display device;

FIG. 16 is a cross-sectional view of another exemplary embodiment of a display device;

FIG. 17 is a cross-sectional view of another exemplary embodiment of a display device;

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FIG. 18 is a cross-sectional view of another exemplary embodiment of a display device;

FIG. 19 is a cross-sectional view of another exemplary embodiment of a single sub-pixel; and

FIG. 20 is a cross-sectional view of another exemplary embodiment of a single sub-pixel.

DETAILED DESCRIPTION

Reference will now be made in detail to exemplary embodiments, examples of which are illustrated in the accompanying drawings, in which like reference numerals refer to like elements throughout. In this regard, the exemplary embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the exemplary embodiments are merely described below, by referring to the figures, to explain exemplary embodiments of the description. Expressions such as "at least one of," when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

As the invention allows for various changes and numerous exemplary embodiments, exemplary embodiments will be illustrated in the drawings and described in detail in the written description. However, this is not intended to limit the invention to particular modes of practice, and it is to be appreciated that all changes, equivalents, and substitutes that do not depart from the spirit and technical scope of the invention are encompassed in the invention. In the description of the invention, certain detailed explanations of the related art are omitted when it is deemed that they may unnecessarily obscure the essence of the invention.

It will be understood that when an element is referred to as being "on" another element, it can be directly on the other element or intervening elements may be therebetween. In contrast, when an element is referred to as being "directly on" another element, there are no intervening elements present.

While such terms as "first," "second," etc., may be used to describe various components, such components must not be limited to the above terms. The above terms are used only to distinguish one component from another.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, the singular forms "a," "an," and "the" are intended to include the plural forms, including "at least one," unless the content clearly indicates otherwise. "Or" means "and/or." As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. It will be further understood that the terms "comprises" and/or "comprising," or "includes" and/or "including" when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof.

Furthermore, relative terms, such as "lower" or "bottom" and "upper" or "top," may be used herein to describe one element's relationship to another element as illustrated in the Figures. It will be understood that relative terms are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures. In an exemplary embodiment, when the device in one of the figures is turned over, elements described as being on the "lower" side of other elements would then be oriented on "upper" sides of the other elements. The exemplary term "lower," can there-

fore, encompasses both an orientation of “lower” and “upper,” depending on the particular orientation of the figure. Similarly, when the device in one of the figures is turned over, elements described as “below” or “beneath” other elements would then be oriented “above” the other elements. The exemplary terms “below” or “beneath” can, therefore, encompass both an orientation of above and below.

“About” or “approximately” as used herein is inclusive of the stated value and means within an acceptable range of deviation for the particular value as determined by one of ordinary skill in the art, considering the measurement in question and the error associated with measurement of the particular quantity (i.e., the limitations of the measurement system). For example, “about” can mean within one or more standard deviations, or within $\pm 30\%$, 20% , 10% , 5% of the stated value.

The terms used in the specification are merely used to describe exemplary embodiments, and are not intended to limit the invention. An expression used in the singular encompasses the expression of the plural, unless it has a clearly different meaning in the context. In the specification, it is to be understood that the terms such as “including,” “having,” and “comprising” are intended to indicate the existence of the features, numbers, steps, actions, components, parts, or combinations thereof disclosed in the specification, and are not intended to preclude the possibility that one or more other features, numbers, steps, actions, components, parts, or combinations thereof may exist or may be added.

The invention will now be described more fully with reference to the accompanying drawings, in which exemplary embodiments of the invention are shown. Like reference numerals in the drawings denote like elements, and thus their repetitive description will be omitted.

FIG. 1 is a perspective view of a display device 100 according to an exemplary embodiment.

Referring to FIG. 1, the display device 100 may include a display panel 110. The display panel 110 may include an active area AA displaying images and an inactive area IAA extending outwardly from the active area AA. According to an exemplary embodiment, the inactive area IAA may surround the active area AA.

According to an exemplary embodiment, the active area AA may extend in a longitudinal direction of the display panel 110, but the invention is not limited thereto. According to an exemplary embodiment, the display panel 110 may be a rigid or flexible panel.

The display device 100 may include a touch screen panel (“TSP”) which recognizes a location where a user contacts the display device 100.

FIG. 2 is a partially exploded perspective view of the TSP 200 according to an exemplary embodiment.

Referring to FIG. 2, the TSP 200 may include a plurality of touch sensor electrodes 230. The plurality of touch sensor electrodes 230 may include a plurality of first touch electrodes 210 and a plurality of second touch electrodes 220.

According to an exemplary embodiment, the TSP 200 is an on-cell TSP in which the plurality of touch sensor electrodes 230 is arranged in the display panel 110 in FIG. 1. However, a structure of the TSP is not limited thereto.

According to another exemplary embodiment, the TSP 200 may have an in-cell TSP structure in which the plurality of touch sensor electrodes 230 are arranged inside the display panel 110 in FIG. 1 or a hybrid TSP structure, which

is a combination of the on-cell TSP and the in-cell TSP structures, for example. Thus, the structure of the TSP is not limited to only one type.

The plurality of first touch electrodes 210 and the plurality of second touch electrodes 220 may be alternately arranged relative to each other on a substrate 201. The substrate 201 may be an encapsulated substrate arranged in the display panel 110 in FIG. 1. The plurality of first touch electrodes 210 and the plurality of second touch electrodes 220 may be arranged in directions crossing each other. Each of the first touch electrodes 210 may be a transmission electrode and each of the second touch electrodes 220 may be a reception electrode.

According to an exemplary embodiment, the TSP 200 may have a mutual-capacitance sensing structure, in which capacitance changes are measured at points where the plurality of touch sensor electrodes 230 cross each other. However, the structure of the TSP 200 is not limited thereto. According to another exemplary embodiment, the structure of the TSP 200 may be a self-capacitance sensing structure, in which the capacitance change is measured at each pixel for touch recognition via a single touch sensor electrode 230. Thus, the structure of the TSP 200 is not limited to one type only.

The plurality of first touch electrodes 210 and the plurality of second touch electrodes 220 may be arranged on the same layer on the substrate 201. In another exemplary embodiment, the plurality of first touch electrodes 210 and the plurality of second touch electrodes 220 may be separated from each other on different layers.

A pair of the first touch electrodes 210 arranged adjacent to each other on the substrate 201 may be electrically connected to each other via a first touch connector 211. A pair of the second touch electrodes 220 arranged adjacent to each other on the substrate 201 may be electrically connected to each other via a second touch connector 221. A pair of the second touch electrodes 220 may be connected to the second touch connector 221 arranged on a different layer via a contact hole 241, to avoid interference with the first touch electrode 210.

An insulating layer 250 covering the plurality of first touch electrodes 210 and the plurality of second touch electrodes 220 may be arranged on the substrate 201.

When an input tool such as a user’s finger or a pen comes close to or contacts the substrate 201, the TSP 200 may detect a touch location by measuring a capacitance change between the first touch electrode 210 and the second touch electrode 220.

Research and development on a display device including a TSP with a plurality of micro-LEDs has been conducted.

FIG. 3 is a plan view of an arrangement of a LED 325 and a touch sensor electrode 326 of a display device 300 according to an exemplary embodiment. FIG. 4 is a cross-sectional view of a single sub-pixel of the display device 300 of FIG. 3.

According to an exemplary embodiment, each of sub-pixels of the display device 300 may include at least one thin film transistor (“TFT”) and at least one micro-LED. However, the TFT is not necessarily feasible only in the structure of FIG. 3, and its number and structure may be variously changeable.

Referring to FIGS. 3 and 4, the display device 300 may include a plurality of sub-pixel areas 301. The sub-pixel areas 301 may be separated from each other in the X-axis and the Y-axis directions on a display substrate 311. The sub-pixel areas 301 may be separated from each other by at least one layer of a bank 323. The sub-pixel area 301 may

include a first area **302** in which the LED **325** is disposed (e.g., placed) and a second area **303** in which a touch sensor electrode **326** is placed.

In an exemplary embodiment, the display substrate **311** may be any one of a rigid glass substrate, a flexible glass substrate, and a flexible polymer substrate, for example. In an exemplary embodiment, the display substrate **311** may be transparent, semi-transparent, or opaque, for example.

A buffer layer **312** may be arranged on the display substrate **311**. The buffer layer **312** may totally cover a top surface of the display substrate **311**. The buffer layer **312** may include an inorganic layer or an organic layer. The buffer layer **312** may be a single layer or a multi-layer.

The TFT may be arranged on the buffer layer **311**. The TFT may include a semiconductor activating layer **313**, a gate electrode **318**, a source electrode **320**, and a drain electrode **321**. According to an exemplary embodiment, the TFT may be a top gate type. However, the invention is not limited thereto, and the TFT may be other types such as a bottom gate type.

The semiconductor activating layer **313** may be arranged on the buffer layer **312**.

The semiconductor activating layer **313** may include a source area **314** and a drain area **315** which are positioned by doping n-type impurity ions or p-type impurity ions. An area between the source area **314** and the drain area **315** may be a channel area **316** in which impurities are not doped. In an exemplary embodiment, the semiconductor activating layer **313** may be an organic semiconductor, an inorganic semiconductor, or amorphous silicon, for example. In another exemplary embodiment, the semiconductor activating layer **313** may be an oxide semiconductor, for example.

A gate insulating layer **317** may be arranged on the semiconductor activating layer **313**. The gate insulating layer **317** may include an inorganic layer. The gate insulating layer **317** may be a single layer or a multi-layer.

The gate electrode **318** may be arranged on the gate insulating layer **317**. The gate electrode **318** may include a material with good conductivity. The gate electrode **318** may be a single layer or a multi-layer.

An interlayer insulating layer **319** may be arranged on the gate electrode **318**. The interlayer insulating layer **319** may include an inorganic layer or an organic layer.

The source electrode **320** and the drain electrode **321** may be arranged on the interlayer insulating layer **319**. In detail, a contact hole may be defined by removing a portion of the gate insulating layer **317** and a portion of the interlayer insulating layer **319**. Then, the source electrode **320** may be electrically connected to the source area **314** via the contact hole and the drain electrode **321** may be electrically connected to the drain area **315** via the contact hole.

A planarization layer **322** may be arranged on the source electrode **320** and the drain electrode **321**. The planarization layer **322** may include an inorganic layer or an organic layer.

At least one layer of the bank **323** separating the sub-pixel areas **301** may be arranged on the planarization layer **322**. The bank **323** may include an inorganic layer or an organic layer. The bank **323** may be transparent or opaque. The bank **323** may include a light absorbing material, a light reflecting material, or a light scattering material. The bank **323** may function as a light blocking layer having low light transmissivity.

An opening **324** may be defined above the TFT by removing a portion of the bank **323**. A first electrode **333** may be arranged on the planarization layer **322** which is exposed by removing the portion of the bank **323**. The first electrode **333** may be electrically connected to the drain

electrode **321** via the contact hole defined in the planarization layer **322**. The first electrode **333** may include a transparent electrode or a metal electrode. The first electrode **333** may have various patterns. In an exemplary embodiment the first electrode **333** may be patterned in an island shape, for example.

The sub-pixel area **301** may include the first area **302** in which the LED **325** is placed and the second area **303** in which the touch sensor electrode **326** is placed. The LED **325** and the touch sensor electrode **326** may be arranged in the sub-pixel area **301**.

In detail, each of the sub-pixel areas **301** may be positioned in the opening **324** surrounded by the bank **323**. The first area **302** in which the LED **325** is placed and the second area **303** in which the touch sensor electrode **326** is placed may be arranged adjacent to each other in each of the sub-pixel areas **301**. According to an exemplary embodiment, a size of the second area **303** may be larger than that of the first area **302**.

The LED **325** may emit light in a certain wavelength band covering a range from ultraviolet ("UV") rays to visible light. In an exemplary embodiment, the LED **325** may be a micro-LED, for example. According to an exemplary embodiment, the LED **325** may be a red-color LED, a green-color LED, a blue-color LED, a white-color LED, or a UV LED, for example.

The LED **325** may include a first contact electrode **328**, a second contact electrode **329**, and a p-n diode **327** arranged between the first contact electrode **328** and the second contact electrode **329**.

The p-n diode **327** may include a p-doped layer **330** on a bottom portion, an n-doped layer **331** on a top portion, and at least one of quantum well layer **332** arranged between the p-doped layer **330** and the n-doped layer **331**. In another exemplary embodiment, the doped layer **331** on the top portion may be the p-doped layer and the doped layer **330** on the bottom portion may be the n-doped layer.

The first contact electrode **328** may be arranged on the p-doped layer **330** on the bottom portion. The first contact electrode **328** may be electrically connected to the first electrode **333**. The second contact electrode **329** may be arranged on the n-doped layer **331** on the top portion. The second contact electrode **329** may be electrically connected to a second electrode **334**. The second electrode **334** may include a transparent electrode or a metal electrode. The second electrode **334** may include various shapes of patterns. According to an exemplary embodiment, the second electrode **334** may be a common electrode.

The touch sensor electrode **326** may be arranged in the second area **303** of each of the sub-pixel areas **301**. The touch sensor electrode **326** may include a metal layer. In a case of a transparent display device, the touch sensor electrode **326** may include a transparent conductive layer such as an indium tin oxide ("ITO") layer.

The touch sensor electrode **326** may be the same as the touch sensor electrode **230** in FIG. 2. The touch sensor electrode **326** may extend to an adjacent sub-pixel area **301**. According to an exemplary embodiment, the touch sensor electrode **326** may be a portion of the first touch electrode **210** or the second touch electrode **220** in FIG. 2.

The touch sensor electrode **326** may be arranged on a same layer as that on which the first electrode **333** is disposed. The touch sensor electrode **326** may include the same material and be obtained via the same process as that of the first electrode **333**. In other exemplary embodiment, the touch sensor electrode **326** may be patterned with another metallic material.

According to an exemplary embodiment, the touch sensor electrode **326** may be electrically connected to the first electrode **333**. In another exemplary embodiment, the touch sensor electrode **326** may apply a separate electrical signal.

The touch sensor electrode **326** may be driven via a mutual-capacitance method or a self-capacitance method, depending on a connection method.

In the case of the mutual-capacitance method, the touch sensor electrode **326** may be an electrode which senses a capacitance change between the plurality of touch electrodes such as the first touch electrode **210** and the second touch electrode **220** in FIG. 2. The plurality of touch electrodes may be arranged on the same layer above the display substrate **311**. In another exemplary embodiment, the plurality of touch electrodes may be separated on different layers above the display substrate **311**.

In the case of the self-capacitance method, the touch sensor electrode **326** may be an electrode which senses a capacitance change of a single touch electrode by using a single touch electrode.

According to an exemplary embodiment, a ground wiring (not illustrated) may be further arranged in the second area **303** of the sub-pixel area **301**.

A filling layer **335** may be filled in the opening **324**. The LED **325** and the touch sensor electrode **326** may be embedded in the filling layer **335**. In an exemplary embodiment, the filling layer **335** may include an organic material, but the invention is not limited thereto.

Likewise, the LED **325** may be arranged in the first area **302** of the sub-pixel area **301**. The touch sensor electrode **326** may be arranged in the second area **303** of the sub-pixel area **301**.

Below, like reference numbers in illustrated drawings above may denote like members performing like functions. Thus, duplicate descriptions will be omitted and only major particular portions of each exemplary embodiment will be selectively described.

FIG. 5 is a plan view of a display device **500** according to another exemplary embodiment.

Referring to FIG. 5, the sub-pixel area **301** may include the first area **302** in which the LED **325** is placed and the second area **303** in which a touch sensor electrode **526** is placed. The filling layer **335** may be filled in the opening **324** from which a portion of the bank **323** is removed.

The LED **325** may be embedded in the filling layer **335**. The touch sensor electrode **526** may be arranged on the filling layer **335**. The touch sensor electrode **526** and the second electrode **334** may be arranged on the same layer. The touch sensor electrode **526** may include the same material and be obtained via the same process as that of the second electrode **334**. The touch sensor electrode **526** may be driven via a mutual-capacitance method or a self-capacitance method, depending on a connection method.

According to an exemplary embodiment, a ground wiring **536** may be further arranged in the second area **303** of the sub-pixel area **301**. The ground wiring **536** and the touch sensor electrode **526** may be arranged on the same layer. The ground wiring **536** may eliminate noise caused by pixel-driving. When the ground wiring **536** is arranged, a capacitance may be reduced, and subsequently, noise may be reduced.

The ground wiring **536** may receive an electrical signal from a power line, through which a constant voltage flows. In another exemplary embodiment, a ground voltage may be applied to the ground wiring **536**. In another exemplary embodiment, the ground wiring **536** may be in a floating state.

FIG. 6 is a plan view of an arrangement of the LED **325** and a touch sensor electrode **626** according to an exemplary embodiment.

Referring to FIG. 6, the LED **325** and the touch sensor electrode **626** of the display device **600** may be arranged in each of sub-pixel areas **601**. Each LED **325** may be arranged in respective sub-pixel areas **601** and the touch sensor electrode **626** may extend to the adjacent sub-pixel area **601**. Each of sub-pixel areas **601** may include a first area **602** in which the LED **325** is placed and a second area **603** in which a touch sensor electrode **626** is placed.

An area of the touch sensor electrode **626** may be expanded by enlarging a gap **g** between a pair of LEDs **325** which are involved in different light-emitting and adjacent to each other in the X-axis direction. When the area of the touch sensor electrode **626** is increased, touch sensitivity may be enhanced.

According to an exemplary embodiment, the LED **325** arranged in each sub-pixel may include at least one of the red-color LED, the green-color LED, the blue-color LED, the white-color LED, and the UV LED. In another exemplary embodiment, a color filter layer having color hue corresponding to respective LEDs **325** may be further arranged above the LED **325**.

FIG. 7 is a plan view of an arrangement of an LED and a touch sensor electrode according to another exemplary embodiment.

Referring to FIG. 7, the display device **700** may include a plurality of sub-pixel areas **701**. The sub-pixel areas **701** may be separated from each other in X-axis and Y-axis directions. The sub-pixel area **701** may include a first area **702** in which the LED **325** is placed and a second area **703** in which a touch sensor electrode **726** is placed. The touch sensor electrode **726** may extend to an adjacent sub-pixel area **701**.

According to an exemplary embodiment, the touch sensor electrode **726** may have a self-capacitance sensing structure. The touch sensor electrode **726** may be an electrode which senses a capacitance change in a single touch electrode. A touch sensor wiring **727** may be arranged on the touch sensor electrode **726**. In an exemplary embodiment a plurality of touch sensor wirings **727** may be arranged in the sub-pixel areas **701** which are arranged along the Y-axis direction, for example.

The touch sensor wiring **727** may extend along the Y-axis direction and may be electrically connected to respective touch sensor electrodes **726** which are continuously arranged in the Y-axis direction. The touch sensor wiring **727** may be electrically connected to the touch sensor electrode **726** and an external device (not illustrated). A changed capacitance may be transferred from the single touch sensor electrode **726** to the external device via the touch sensor wiring **727**, and a sensor voltage generated by the external device may be transferred to the touch sensor electrode **726**.

According to an exemplary embodiment, the touch sensor wiring **727** may be directly or indirectly connected to the touch sensor electrode **726** on the touch sensor electrode **726**.

Referring to FIG. 8, the sub-pixel area **301** of the display device **800** may include the first area **302** in which the LED **325** is placed and the second area **303** in which a touch sensor electrode **826** is placed. The filling layer **335** may be filled in the opening **324** which is defined by removing the portion of the bank **323**.

The touch sensor electrode **826** and the first electrode **333** may be arranged on the same layer. A touch sensor wiring

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827 may be arranged on the filling layer 335. The touch sensor wiring 827 may be arranged in the second area 303. The touch sensor wiring 827 may be electrically connected to the touch sensor electrode 826 via a contact hole 828 defined in the filling layer 335.

According to an exemplary embodiment, the touch sensor wiring 827 and at least one of the gate electrode 318, the source electrode 320, and the drain electrode 321 may be included on the same layer. However, the current exemplary embodiment is not limited to a single location.

Referring to FIG. 9, the sub-pixel area 301 of the display device 900 may include the first area 302 in which the LED 325 is placed and the second area 303 in which a touch sensor electrode 926 is placed. The filling layer 335 may be filled in the opening 324 which is defined by removing the portion of the bank 323.

The touch sensor electrode 926 may be arranged on the filling layer 335. The touch sensor electrode 926 and the second electrode 334 may be arranged on the same layer. A touch sensor wiring 927 may be arranged on the touch sensor electrode 926. The touch sensor wiring 927 may be directly connected to the touch sensor electrode 926.

FIG. 10 is a plan view of an arrangement of the LED 325 and a touch sensor electrode 1026 according to another exemplary embodiment.

Referring to FIG. 10, a display device 1000 may include a plurality of sub-pixel areas 1001. The sub-pixel areas 1001 may be separated from each other in X-axis and the Y-axis directions. The sub-pixel area 1001 may include a first area 1002 in which the LED 325 is placed and a second area 1003 in which the touch sensor electrode 1026 is placed. The touch sensor electrode 1026 may extend to an adjacent sub-pixel area 1001.

According to an exemplary embodiment, the touch sensor electrode 1026 may have a mutual-capacitance sensing structure. The touch sensor electrode 1026 may be an electrode which senses the capacitance change generated between a first touch electrode 1028 and a second touch electrode 1029. The first touch electrode 1028 and the second touch electrode 1029 may be arranged on the same layer.

The first touch electrode 1028 and the second touch electrode 1029 may be alternately arranged in the X-axis direction. The pair of first touch electrodes 1028, which are separated from each other with the second touch electrode 1029 therebetween in the X-axis direction, may be electrically connected to each other via a connecting wiring 1027 which is arranged on a different layer with respect to the first touch electrode 1028. According to an exemplary embodiment, the connecting wiring 1027 may include the same material as that of at least one of the gate electrode, the source electrode, and the drain electrode. In another exemplary embodiment, the connecting wiring 1027 may be electrically connected to any one of the gate electrode, the source electrode, and the drain electrode.

The second touch electrodes 1029 may be respectively connected to separate touch sensor wirings, or may be directly connected to a touch integrated circuit ("IC") on one edge of the display device 1000.

FIG. 11 is a plan view of an arrangement of an LED and a touch sensor electrode according to another exemplary embodiment, and FIG. 12 is a cross-sectional view of a single sub-pixel in FIG. 11.

Referring to FIGS. 11 and 12, the display device 1100 may include a plurality of sub-pixel areas 1101. The sub-pixel areas 1101 may be separated from each other in the X-axis and the Y-axis directions. The sub-pixel area 1101

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may include a first area 1102 in which the LED 325 is placed and a second area 1103 in which a touch sensor electrode 1126 is placed. The touch sensor electrode 1126 may extend to an adjacent sub-pixel area 1101.

According to an exemplary embodiment, the touch sensor electrode 1126 may have the mutual-capacitance sensing structure. The touch sensor electrode 1126 may include a first touch electrode 1128 and a second touch electrode 1129. The first touch electrode 1128 and the second touch electrode 1129 may be alternately arranged in the X-axis direction. Unlike as illustrated in FIG. 10, the first touch electrode 1128 and the second touch electrode 1129 may be separated from each other on different layers.

The first touch electrode 1128, which may be a metal electrode, and a single electrode arranged on the TFT may be a metal electrode arranged on the same layer. In an exemplary embodiment, the first touch electrode 1128 and the source electrode 320 or the drain electrode 321 may be arranged on the same layer, for example. The first touch electrode 1128 may include the same material in the same process as that of the source electrode 320 or the drain electrode 321. In another exemplary embodiment, the first touch electrode 1128, which may be the metal electrode, and the gate electrode 318 may be arranged on the same layer.

The second touch electrode 1129 and the first electrode 333 may be arranged on the same layer. The second touch electrode 1129 may include the same material in the same process as that of the first electrode 333. In another exemplary embodiment, the second touch electrode 1129 and the second electrode 334 may be arranged on the same layer and may include the same material in the same process.

In another exemplary embodiment, the first touch electrode 1128 may include a separate metal layer, for example, a transparent conductive layer, and the second touch electrode 1129 and at least one of the gate electrode 318, the source electrode 320, the drain electrode 321, the first electrode 333, and the second electrode 334 may be arranged on the same layer.

FIG. 13 is a cross-sectional view of a display device 1300 according to another exemplary embodiment.

Referring to FIG. 13, a first function layer 1301 encapsulating the sub-pixel area 301 may be arranged on the display substrate 311. According to an exemplary embodiment, the first function layer 1301 may be an encapsulating layer, but it is not limited thereto. A second function layer 1303 may be arranged above the first function layer 1301 with a medium layer 1302 therebetween. The second function layer 1303 may be a window cover, but it is not limited thereto.

The medium layer 1302 may receive pressure caused by a user's contact and may be a material with a cushion function. According to an exemplary embodiment, the medium layer 1302 may be an air layer, but it is not limited thereto. A spacer 1304 maintaining a cell gap may be arranged between the first function layer 1301 and the second function layer 1303. A sealant 1305 may be applied to edges where the first function layer 1301 and the second function layer 1303 may face.

The sub-pixel area 301 may include the first area 302 in which the LED 325 is placed and the second area 303 in which the touch sensor electrode 326 is placed. The touch sensor electrode 326 and the first electrode 333 may be arranged on the same layer. In another exemplary embodiment, the touch sensor electrode 326 and the second electrode 334 may be arranged on the same layer.

The touch sensor electrode 326 may correspond to an electrode which senses the capacitance change in the X-axis

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and the Y-axis crossing the X-axis. A force sensing electrode **1306**, which forms capacitance with the touch sensor electrode **326** in a Z-axis perpendicular to the X-axis and the Y-axis and senses pressure in accordance with the capacitance change, may be further arranged on the display substrate **311**. According to an exemplary embodiment, the force sensing electrode **1306** may be arranged on one surface of the second function layer **1303** facing the first function layer **1301**.

Since capacitance is generated between the touch sensor electrode **326** and the force sensing electrode **1306**, sensing the capacitance change in the Z-axis may be possible. In detail, when an input tool is pressed, an applied force may be sensed depending on the capacitance change between the touch sensor electrode **326** and the force sensor electrode **1306**.

The force sensor electrode **1306** may be patterned with a particular pattern on the second function layer **1303**. In detail, the force sensor electrode **1306** may be patterned with a plurality of patterns having different areas from each other, to sense each location in accordance with the force applied to the display device **1300**. In another exemplary embodiment, the force sensor electrode **1306** may be entirely disposed on the second function layer **1303**, to sense the force only.

In an exemplary embodiment, the force sensor electrode **1306** may include conductive polymer materials such as poly 3, 4-ethylenedioxy thiophene ("PEDOT"), polyacetylene, and polypyrrole. Since the force sensor electrode **1306** has resistance per unit area more than 100 times larger than that of the transparent conductive layer such as an ITO layer, while maintaining electrical conductivity, the force sensor electrode **1306** may be applicable to an electrode of the TSP.

A connector **1307** which is electrically connected to the force sensor electrode **1306** may be arranged between the first function layer **1301** and the second function layer **1303**. The connector **1307** may be connected to a pad on the display substrate **311** via the contact hole **1308** defined in the first function layer **1301** and may transfer a force sensor signal to the external device.

Likewise, the structure, in which locations in the X-axis and the Y-axis are sensed by using the touch sensor electrode **326** and a force in the Z-axis is sensed by using the force sensor electrode **1306**, may be applied to various display devices.

Referring to FIG. **14**, a display device **1400** may be an organic light-emitting display device. In detail, a display unit **1402** including the TFT and an organic LED may be arranged on a display substrate **1401**, a touch sensor electrode **1403** may be arranged on the display unit **1402**, and a first function layer **1404** corresponding to encapsulation may be arranged on the touch sensor electrode **1403**. A second function layer **1406**, which corresponds to the window cover, may be arranged above the first function layer **1404** with a medium layer **1405** therebetween. A force sensor electrode **1407** may be arranged on one surface of the second function layer **1406** facing the first function layer **1404**.

The touch sensor electrode **1403** may be connected to a pad on the display substrate **1401** via a contact hole **1408** defined in the display unit **1402** and may transfer the sensor signal to the external device. In addition, the force sensor electrode **1407** may be connected to a connector **1409** and may transfer the force sensor signal to the external device.

The structure described above may sense the force applied in accordance with the capacitance change between the touch sensor electrode **1403** and the force sensor electrode **1407**.

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Referring to FIG. **15**, a display device **1500** may be an organic light-emitting display device. In detail, a display unit **1502** including the TFT and the organic LED may be arranged on the display substrate **1501**. Unlike as illustrated in FIG. **14**, a first function layer **1504** corresponding to encapsulation may be arranged on the display unit **1502**. A touch sensor electrode **1503** may be arranged on the first function layer **1504**. A second function layer **1506**, which corresponds to the window cover, may be arranged above the touch sensor electrode **1503** with a medium layer **1505** therebetween. A force sensor electrode **1507** may be arranged on one surface of the second function layer **1506**. The force sensor electrode **1507** may be connected to a connector **1509** and may transfer the force sensor signal to the external device.

The structure described above may sense the force applied in accordance with the capacitance change between the touch sensor electrode **1503** and the force sensor electrode **1507**.

Referring to FIG. **16**, a display device **1600** may be a liquid crystal display device. In detail, a display unit **1602** including a crystal display element may be arranged on the display substrate **1601**, and a touch sensor electrode **1603** may be arranged on the display unit **1602**. A function layer **1604**, which corresponds to a color filter substrate, may be arranged on the touch sensor electrode **1603**, and a force sensor electrode **1605** may be arranged on the function layer **1604**. The force sensor electrode **1605** may be connected to a connector **1606** and may transfer the force sensor signal to the external device.

The structure described above may sense the applied force via the capacitance change between the touch sensor electrode **1603** and the force sensor electrode **1605**.

Referring to FIG. **17**, a display device **1700** may be a liquid crystal display device. In detail, a display unit **1702** including a crystal display element may be arranged on a display substrate **1701**. Unlike as illustrated in FIG. **16**, a first function layer **1704**, which corresponds to a color filter substrate, may be arranged on the display unit **1702** and a touch sensor electrode **1703** may be arranged on the first function layer **1704**. A second function layer **1708** such as the window cover may be arranged above the touch sensor electrode **1703** with a medium layer **1707** therebetween. A force sensor electrode **1705** may be arranged on one surface of the second function layer **1708**. The force sensor electrode **1705** may be connected to a connector **1706** and may transfer the force sensor signal to the external device.

The structure described above may sense the applied force via the capacitance change between the touch sensor electrode **1703** and the force sensor electrode **1705**.

Referring to FIG. **18**, a display device **1800** may include a display substrate **1801**. A display unit **1802** may be arranged on the display substrate **1801**. The display unit **1802** may include a micro-LED. A touch sensor electrode **1803** may be arranged on the display unit **1802**. In another exemplary embodiment, the display unit **1802** may include a LCD element, or an organic LED. In another exemplary embodiment, a function layer such as a color filter substrate or an encapsulation layer may be arranged on the display **1802**.

A function layer **1809**, which corresponds to the window cover, may be arranged above the touch sensor electrode **1803** with a medium layer **1810** therebetween. A force sensor electrode **1805** may be arranged on a function layer **1809** facing the touch sensor electrode **1803**.

According to an exemplary embodiment, the medium layer **1810** may include an air layer. A refractive index

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matching layer ("RIML") **1806** may be arranged on at least one surface of the force sensor electrode **1805** to improve reflective index affected by the air layer. The RIML **1806** may include a first RIML **1807** arranged on one surface of the force sensor electrode **1805** facing the touch sensor electrode **1803** and a second RIML **1808** arranged on the other surface of the force sensor electrode **1805** facing the function layer **1809**.

In an exemplary embodiment, the RIML **1806** may include silicon oxide (SiO₂) or silicon nitride (SiN_x), for example. The refractive index of the first RIML **1807** and that of the second RIML **1808** may be less than that of the force sensor electrode **1805**. The RIML **1806** with a low refractive index and the force sensor electrode **1805** with a high refractive index are alternately arranged on the medium layer **1810**. Thus, the refractive index of the display device **1800** may be improved and poor show-through of the touch sensor electrode **1803** may be improved.

FIG. **19** is a cross-sectional view of a single sub-pixel according to another exemplary embodiment.

Referring to FIG. **19**, the display device **1900** may include the plurality of sub-pixel areas **301**. The sub-pixel area **301** may include the first area **302** in which the LED **325** is placed and the second area **303** in which the touch sensor electrode **1926** is placed. The touch sensor electrode **1926** may extend to an adjacent sub-pixel area **301**.

According to an exemplary embodiment, the touch sensor electrode **1926** may include a first touch electrode **1927** and a second touch electrode **1928**. The first touch electrode **1927** and the first electrode **333** may be arranged on the same layer. In another exemplary embodiment, the first touch electrode **1927** and at least one of the gate electrode **318**, the source electrode **320**, and the drain electrode **321**, which are arranged on the TFT, may be arranged on the same layer.

A first bank **1903** may be arranged on the circumference of the sub-pixel area **301**. The LED **325** and the first touch electrode **1927** may be arranged in the opening **324** with a portion of the first bank **1903** removed therefrom.

A second bank **1904** may be further arranged on the first bank **1903**. The second bank **1904** may embed the LED **325** and the first touch electrode **1927**. The second bank **1904** may planarize the top surface of the LED **325**.

The second electrode **334** may be electrically connected to the LED **325** on the second bank **1904**. The second touch electrode **1928** and the second electrode **333** may be arranged on the same layer. The touch sensor electrode **1926** may correspond to an electrode which senses the capacitance change between the first touch electrode **1927** and the second touch electrode **1928**.

A color filter **1901** may be arranged over the LED **325**. The color filter **1901** may transform light emitted from the LED **325** or increase color purity.

A black matrix **1902** may be arranged on the circumference of the color filter **1901**. The black matrix **1902** may surround the circumference of the LED **325**. In another exemplary embodiment, the black matrix **1902** may be arranged between adjacent sub-pixel areas **301**. Since the color filter **1901** and the black matrix **1902** are arranged, a polarization plate is not needed and reflection of external light may be improved.

An encapsulation layer **1905** may be arranged on the outermost circumference of the display substrate **311** to protect each element arranged on the display substrate **311**. The encapsulation layer **1905** may include a lamination of at least one of inorganic materials and at least one of organic

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materials. In another exemplary embodiment, the encapsulation layer **1905** may include an inorganic material.

FIG. **20** is a cross-sectional view of a single sub-pixel according to another exemplary embodiment.

Referring to FIG. **20**, the display device **2000** may include the plurality of sub-pixel areas **301**. The sub-pixel area **301** may include the first area **302** in which the LED **325** is placed and the second area **303** in which a touch sensor electrode **2026** is placed. The touch sensor electrode **2026** may extend to an adjacent sub-pixel area **301**. According to an exemplary embodiment, the touch sensor electrode **2026** may include a first touch electrode **2027** and a second touch electrode **2028**. The first touch electrode **2027** and the first electrode **333** may be arranged on the same layer. In another exemplary embodiment, the first touch electrode **2027** and at least one of the gate electrode **318**, the source electrode **320**, and the drain electrode **321**, which are provided in the TFT, may be arranged on the same layer.

A first bank **2003** may be arranged on the circumference of the sub-pixel area **301**. The first bank **2003** may extend to an adjacent sub-pixel area **301**. The LED **325** and the first touch electrode **2027** may be arranged in the opening **324** which is provided by removing the portion of the first bank **2003**.

A second bank **2004** may be further arranged on the first bank **2003**. The second bank **2004** may embed the LED **325** and the first touch electrode **2027**. The second bank **2004** may be independently arranged on respective sub-pixel areas **301** such that the LEDs **325** are embedded.

According to an exemplary embodiment, the second bank **2004** may extend from the first area **302** in which the LED **325** is placed to the second area **302** in which the touch sensor electrode **2026** is placed. In another exemplary embodiment, the second bank **2004** may be arranged on only the first area **302** including the LED **325**.

According to an exemplary embodiment, the second bank **2004** may include a scattering material or a color conversion material.

The second electrode **334** may be electrically connected to the LED **325** on the second bank **2004**. A second touch electrode **2028** and the second electrode **334** may be arranged on the same layer. The second touch electrode **2028** may correspond to an electrode which senses the capacitance change between the first touch electrode **2027** and the second touch electrode **2028**.

A color filter **2001** may be arranged above the LED **325**. The color filter **2001** may transform light emitted from the LED **325** or increase color purity.

A black matrix **2002** may be arranged on the circumference of the color filter **2001**. According to an exemplary embodiment, the black matrix **2002** may surround the circumference of the LED **325**. In another exemplary embodiment, the black matrix **2002** may be arranged between adjacent sub-pixel areas **301**.

An encapsulation layer **2005** may be arranged on the outermost circumference of the display substrate **311** to protect each element arranged on the display substrate **311**. The encapsulation layer **2005** may include a lamination of at least one of inorganic materials and at least one of organic materials.

It should be understood that the exemplary embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or exemplary embodiments within each exemplary embodiment should typically be considered as available for other similar features or exemplary embodiments in other exemplary embodiments.

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While one or more exemplary embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope as defined by the following claims.

What is claimed is:

1. An organic light emitting diode ("LED") device, comprising:

a substrate;

a display unit comprising a plurality of thin film transistors ("TFT"s) and a plurality of organic LEDs on the substrate;

a pad on the substrate;

an encapsulation layer disposed on the substrate and comprising at least one inorganic layer and at least one organic layer;

a touch sensor disposed on the encapsulation layer and comprising connecting wires, first sensor electrodes and second sensor electrodes, the second sensor electrodes being spaced apart from the first sensor electrodes in a plan view; and

a window cover above the touch sensor,

wherein the touch sensor is electrically connected to the pad via a contact hole in the display unit,

wherein one of the first sensor electrodes comprises a first portion and a second portion, the first portion mainly extending in a first direction in a plan view, the second portion mainly extending in a second direction intersecting the first direction in a plan view, and the first portion and the second portion being directly connected to each other at a point where the first portion and the second portion intersect,

wherein one of the second sensor electrodes comprises a third portion and a fourth portion, the third portion mainly extending in a first direction in a plan view and the fourth portion mainly extending in the second direction in a plan view,

wherein the second portion of the one first sensor electrode is disposed between a pair of organic LEDs directly next to each other in the first direction in a plan view,

wherein one of the pair of organic LEDs comprises a first side and a second side opposite to the first side, the first side facing the second portion of the one first sensor electrode in a plan view, and the second side facing the fourth portion of the one second sensor electrode in a plan view,

wherein the TFT comprises an active layer, a gate electrode, a source electrode, and a drain electrode,

wherein the first sensor electrodes, the connecting wires, the source electrode, and the drain electrode comprise metallic materials, and

wherein the first sensor electrodes and the source electrode, or the first sensor electrodes and the drain electrode comprise same materials.

2. The organic LED device of claim 1, wherein the touch sensor is directly on the encapsulation layer.

3. The organic LED device of claim 1, further comprising a dummy pattern, wherein the dummy pattern, the first sensor electrodes and the second sensor electrodes are disposed on a same layer, and wherein the dummy pattern, the first sensor electrodes, the source electrode and the drain electrode comprise same materials.

4. The organic LED device of claim 1, further comprising color filters on the organic LEDs, wherein the color filters have color hues respectively corresponding to the organic LEDs.

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5. The organic LED device of claim 4, further comprising a bank surrounding the organic LEDs, wherein the bank is opaque.

6. The organic LED device of claim 5, further comprising a black matrix,

wherein the black matrix is arranged on a circumference of one of the color filters,

wherein a side surface of the black matrix directly contacts a side surface of the one color filter,

wherein the black matrix is disposed on the touch sensor, and

wherein the black matrix is overlapped the touch sensor in a plan view.

7. The organic LED device of claim 1, wherein the one of the pair of organic LEDs is directly next to the fourth portion of the one second sensor electrode, and

wherein the third portion and the fourth portion of the one second sensor electrode are directly connected to each other at a point where the third portion and the fourth portion intersect.

8. An organic light emitting diode ("LED") device, comprising:

a substrate;

a display unit comprising a plurality of thin film transistors ("TFT"s) and a plurality of organic LEDs on the substrate;

a pad on the substrate,

a touch sensor on the substrate and comprising connecting wires, first sensor electrodes and second sensor electrodes, the second sensor electrodes being spaced apart from the first sensor electrodes in a plan view;

color filters on the touch sensor; and

a window cover on the color filters,

wherein the touch sensor is electrically connected to the pad via a contact hole in the display unit,

wherein one of the first sensor electrodes comprises a first portion and a second portion, the first portion mainly extending in a first direction in a plan view, the second portion mainly extending in a second direction intersecting the first direction in a plan view, and the first portion and the second portion being directly connected to each other at a point where the first portion and the second portion intersect,

wherein one of the second sensor electrodes comprises a third portion and a fourth portion, the third portion mainly extending in a first direction in a plan view and the fourth portion mainly extending in the second direction in a plan view,

wherein the second portion of the one first sensor electrode is disposed between a pair of organic LEDs directly next to each other in the first direction in a plan view,

wherein one of the pair of organic LEDs comprises a first side and a second side opposite to the first side, the first side facing the second portion of the one first sensor electrode in a plan view, and the second side facing the fourth portion of the one second sensor electrode in a plan view,

wherein the TFT comprises an active layer, a gate electrode, a source electrode, and a drain electrode,

wherein the first sensor electrodes, the connecting wires, the source electrode, and the drain electrode comprise metallic materials,

wherein the first sensor electrodes and the source electrode, or the first sensor electrode and the drain electrode comprise same materials, and

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wherein the color filters have color hues respectively corresponding to the organic LEDs.

9. The organic LED device of claim 8, further comprising a bank surrounding the organic LEDs, wherein the bank is opaque.

10. The organic LED device of claim 9, further comprising a black matrix, wherein the black matrix is arranged on a circumference of one of the color filters, wherein a side surface of the black matrix directly contacts a side surface of the one color filter, wherein the black matrix is disposed on the touch sensor, and wherein the black matrix is overlapped the touch sensor in a plan view.

11. The organic LED device of claim 10, wherein the black matrix is disposed on the touch sensor, and wherein the black matrix is overlapped the touch sensor in a plan view.

12. The organic LED device of claim 8, further comprising an encapsulation layer disposed on the display unit and comprising at least one inorganic layers and at least one organic layers.

13. The organic LED device of claim 12, wherein the touch sensor is directly on the encapsulation layer.

14. The organic LED device of claim 8, further comprising a dummy pattern, wherein the dummy pattern, the first sensor electrodes and the second sensor electrodes are disposed on a same layer, and wherein the dummy pattern, the first sensor electrodes, the source electrode and the drain electrode comprise same materials.

15. The organic LED device of claim 8, wherein the one of the pair of organic LEDs is directly next to the fourth portion of the one second sensor electrode, and wherein the third portion and the fourth portion of the one second sensor electrode are directly connected to each other at a point where the third portion and the fourth portion intersect.

16. A mobile device with display, comprising:
a substrate;
a display unit comprising a plurality of thin film transistors ("TFT"s) and a plurality of organic LEDs on the substrate;
a pad on the substrate;
an encapsulation layer on the display unit and comprising at least one inorganic layer and at least one organic layer;
a touch sensor on the encapsulation layer and comprising connecting wires, first sensor electrodes and second sensor electrodes, the second sense electrodes being spaced apart from the first sensor electrodes in a plan view; and
a window cover above the touch sensor, wherein the touch sensor is electrically connected to the pad via a contact hole in the display unit, wherein one of the first sensor electrodes comprises a first portion and a second portion, the first portion mainly extending in a first direction in a plan view, the second portion mainly extending in a second direction intersecting the first direction in a plan view, and the first portion and the second portion being directly connected to each other at a point where the first portion and the second portion intersect,

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wherein one of the second sensor electrodes comprises a third portion and a fourth portion, the third portion mainly extending in a first direction in a plan view and the fourth portion mainly extending in the second direction in a plan view,

wherein the second portion of the one first sensor electrode is disposed between a pair of organic LEDs directly next to each other in the first direction in a plan view,

wherein one of the pair of organic LEDs comprises a first side and a second side opposite to the first side, the first side facing the second portion of the one first sensor electrode in a plan view, and the second side facing the fourth portion of the one second sensor electrode in a plan view,

wherein the TFT comprises an active layer, a gate electrode, a source electrode, and a drain electrode, wherein the first sensor electrodes, the connecting wires, the source electrode, and the drain electrode comprise metallic materials,

wherein the first sensor electrodes and the source electrode, or the first sensor electrode and the drain electrode comprise same materials, and

wherein the mobile device comprises a smart phone, a lap top computer, a digital camera, a camcorder, a mobile information terminal, a tablet personal computer, a watch, a desk top computer, a television ("TV") set, an outdoor advertisement board, an exhibition display device, an automobile instrument panel or a head up display ("HUD").

17. The mobile device of claim 16, wherein the touch sensor is directly on the encapsulation layer,

wherein the one of the pair of organic LEDs is directly next to the fourth portion of the one second sensor electrode, and

wherein the third portion and the fourth portion of the one second sensor electrode are directly connected to each other at a point where the third portion and the fourth portion intersect.

18. The mobile device of claim 16, further comprising a dummy pattern,

wherein the dummy pattern, the first sensor electrodes and the second sensor electrodes are disposed on a same layer, and

wherein the dummy pattern, the first sensor electrodes, the source electrode and the drain electrode comprise same materials.

19. The mobile device of claim 16, further comprising: color filters on the organic LEDs, wherein the color filters have color hues respectively corresponding to the organic LEDs; and

a bank surrounding an organic LED, wherein the bank is opaque.

20. The mobile device of claim 19, further comprising a black matrix,

wherein the black matrix is arranged on a circumference of one of the color filters,

wherein a side surface of the black matrix directly contacts a side surface of the one color filter,

wherein the black matrix is disposed on the touch sensor, and

wherein the black matrix is overlapped the touch sensor in a plan view.

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